Assessing the Heterogeneous Impact of Economy-Wide Shocks: A Causal Machine Learning Approach Applied to Colombian Firms*

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Abstract

Our paper presents a causal Machine Learning (ML) methodology to study the heterogeneous effects of economy-wide shocks and applies it to the impact of the COVID-19 crisis on exports. This method is applicable in scenarios where, due to the pervasive nature of the shock, it is difficult to identify a control group that is not affected by the shock and to determine ex ante differences in shock intensity across units. In particular, our study investigates the effectiveness of different machine learning techniques in predicting firms' trade and, by building on recent developments in causal ML, these predictions are used to reconstruct the counterfactual distribution of firms' trade under different COVID-19 scenarios and investigate the heterogeneity of treatment effects. Specifically, we focus on the probability of Colombian firms surviving in the export market under two different scenarios: a COVID-19 setting and a non-COVID-19 counterfactual situation. On average, we find that the COVID-19 shock decreased a firm's probability of surviving in the export market by about 20 percentage points in April 2020. We study the treatment effect heterogeneity by employing a classification analysis that compares the characteristics of the firms on the tails of the estimated distribution of the individual treatment effects.

Keywords: Economy-wide shocks; Causal Machine Learning; Heterogeneous Treatment Effects; COVID-19; International Trade.

JEL Codes: F14; F17; D22; L25

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1 Introduction

This paper presents a methodology to study the heterogeneous effects of economy-wide shocks, applicable in scenarios where neither a control group unaffected by the shock nor an ex-ante definition of the intensity of the shock for each unit is possible. We apply this methodology to the impact of the COVID-19 crisis on exports. In particular, we aim to estimate the causal effect of COVID-19 on a firm's probability of survival in the export markets and to study the heterogeneity of this effect. The main hurdles for this evaluation task are related to the pervasiveness of the COVID-19 shock. On the one hand, the fact that all firms are eventually exposed to the effects of the COVID-19 crisis makes it hardly possible to find a control group of firms to be used to build a counterfactual non-COVID-19 scenario. On the other hand, adopting a continuous treatment variable would imply defining ex-ante the main patterns through which the COVID-19 shock has affected firm-level trade. This task is highly demanding, given that the economy-wide impact of the shock is coupled with complex interdependencies between firms and products across sectors and countries. The paper's main idea is to address these evaluation challenges, which are present when studying the heterogeneous impact of economy-wide shocks, by leveraging the predictive capabilities of Machine Learning (ML) techniques.

To face the COVID-19 crisis, governments implemented social distancing and lockdown policies, exacerbating supply and demand shocks (World Bank, 2020). In a interconnected world, the impact of the pandemic on international trade has gained great attention (Felbermayr and Görg, 2020; Antràs et al., 2023; Bonadio et al., 2020; Evenett, 2020).

Global trade, which is typically more volatile than output and tends to fall sharply during a crisis, has shown the biggest fall since the 2009 global financial crisis. From the beginning of the COVID-19 epidemic, scholars underlined that, though its impact on international trade could have been comparable to the Great Trade Collapse of 2008-2009, this time, the demand-side shock is accompanied by a supply-side shock (Baldwin and Tomiura, 2020). Moreover, this supply-side effect could be reinforced by a supply-side contagion via importing/supply chains, which have grown in relevance during the last decade (Antras and Chor, 2022). In other words, supply disruptions in the countries providing intermediate inputs to a given country are likely to hurt also its export performance (Halpern et al., 2015; Navas et al., 2020).

We focus on Colombian exporters because of the vulnerability of the Colombian economy to the COVID-19 shock and the availability of detailed customs data. As in many other developing and industrialized countries, Colombia experienced domestic supply and demand shocks in 2020, with factory closures, the suspension of some public services and supply chain disruptions.

By interpreting exporters' dynamics as a complex learning process,[1]

[1]Firms have heterogeneous and incomplete information about the trade opportunities. This is true both on the exporting and the importing side of firm activities. For example, in Albornoz et al. (2012) and Eslava

this paper's first contribution is exploring and comparing the effectiveness of different ML techniques in predicting firms' trade status in two different scenarios, a COVID-19 and a non-COVID-19 setting. ML techniques have been successfully applied to predict firm performances in high-dimensional contexts (Bargagli-Stoffi et al., 2021) in which the number of potentially relevant explanatory variables is very high.

Our paper fits into a nascent literature that is applying ML techniques to study international trade patterns (Breinlich et al., 2022) and, up to what we know, in our study for the first time ML techniques are used to investigate firm-level international trade performance. Estimating more accurately the likelihood of a firm's success in exporting could be useful to increase the effectiveness of export promotion agencies (Van Biesebroeck et al., 2015) by helping them target their activities. However, the effectiveness of ML in improving the prediction of a firm's success cannot be taken for granted, especially for developing countries, as shown by McKenzie and Sansone (2019).

This paper's second and main contribution is to show how to use these predictions to estimate the causal effect of the COVID-19 shock at the firm level and to study its possible heterogeneity. We use the estimated ML model with the best performance in predicting the 2019 export status of firms exporting in 2018 to build a 2020 non-COVID-19 counterfactual outcome for firms exporting in 2019. Then, we compare these counterfactual non-COVID-19 firm-level export probabilities with the predicted probabilities of the best-performing ML model using the characteristics of 2019 exporters to predict their export status in 2020. The latter estimated probabilities summarize the information on the observed COVID-19 scenario and express it in a metric comparable with the estimated counterfactual non-COVID-19 outcomes. In the literature using ML counterfactuals when no control group is available (Cerqua and Letta, 2020; Fabra et al., 2022), it is instead common to estimate causal effects by comparing the counterfactual predictions with the observed outcome in case of treatment, following the so-called "consistency assumption": if the outcome in case of treatment is observed then it also represents the potential outcome under treatment. We follow Chernozhukov et al. (2023)[2] by using ML techniques to reconstruct firm potential outcomes in the case of no treatment and also to predict the outcomes in the treatment scenario. From a methodological standpoint, our study represents the pioneering application and adaptation of the generic machine learning tools proposed by Chernozhukov et al. (2023) in a context where a control group is unavailable.[3] Furthermore, we provide guidance on utilizing in-time placebo tests to assess the credibility of counterfactual estimates. Additionally, we compare the estimation results of the average treatment effect and treatment effect heterogeneity

et al. (2015) exporting firms are uncertain and learn about the appeal of their products and, more in general, about the profitability of exporting their products on the international markets. By searching for clients and observing their realized profitability, firms update their beliefs about their capabilities in international markets.

[2]See formulas 2.6 and 2.7 in Chernozhukov et al. (2023).

[3]For applications of the generic machine learning methodology in economics see Deryugina et al. (2019); Magnan et al. (2021); Baiardi and Naghi (2024); Buhl-Wiggers et al. (2024).

obtained by employing the predicted outcomes in the case of treatment, as proposed by Chernozhukov et al. (2023), with those obtained using the observed outcomes of treated units (i.e., following the Cerqua and Letta (2020) and Fabra et al. (2022) approach). Our findings suggest that while the estimates of the average treatment effect remain robust across methodologies, the former approach should be preferred when the objective is to identify the observations with the highest and lowest treatment effects, and subsequently determine the factors contributing to treatment effect heterogeneity.

Examining the heterogeneous effects of economy-wide shocks is a crucial undertaking as it represents the foundational stage in devising policy interventions intended to mitigate their deleterious outcomes and reactivate economic operations.

However, from a methodological point of view, investigating the treatment effect heterogeneity is not a straightforward task when its potential determinants are many. The traditional approach splits the sample into groups to assess the significance of the difference in the treatment effects of the groups. Unfortunately, this approach is prone to overfitting, and finding statistically significant differences out of all possible splits might be entirely due to random noise. Recently, new tools based on ML have been developed to identify subgroups that are particularly responsive to the treatment (Athey et al., 2019; Chernozhukov et al., 2023). Building on the recent progress in causal ML application to the analysis of heterogeneous effects, in this paper we adopt an agnostic ML model to investigate treatment effect heterogeneity. In particular, we interpret the estimated effects stemming from our ML counterfactual empirical model by using the Sorted Effects method (Chernozhukov et al., 2018, 2023). This method focuses on the tails of the estimated distribution of the firm-level treatment effect to identify the units that are most affected and those that are least affected by the treatment (whose characteristics are compared). We provide evidence that contrasting the estimated counterfactual outcomes with the outcomes predicted for the treatment scenario (and not directly with the observed outcomes under treatment) is crucial to correct the estimation error arising from the imperfect reconstruction of the unobservable counterfactual.

Our paper is connected to the literature on the heterogeneous impact of the COVID-19 shock on trade. Using firm-level monthly data on Spanish trade in goods, de Lucio et al. (2020) find that exports decreased more in countries that introduced strict policies to contain COVID-19 and for goods that are consumed outside the household, particularly between March and May, showing how Spain's export performance during the pandemic depends on COVID-19-induced demand shocks in export markets and the characteristics of products. Using monthly bilateral product-level trade flows that cover three-quarters of world trade, Berthou and Stumpner (2024) also find that the impact of the COVID-19 shock on exports was particularly strong in the spring of 2020, and that demand shocks related to COVID-19 impacted exports directly (shocks in importing countries) but also indirectly (shocks in third countries). Using a sector-level gravity model, Espitia et al. (2021) show that, during the

COVID-19 crisis, sectors that tend to be relatively less internationally integrated suffered less from foreign shocks but were more vulnerable to domestic shocks. Using data on Chinese imports at the country-product level, also Liu et al. (2021) show that the COVID-19 effects are heterogeneous, being weaker for medical goods and stronger for durable consumption goods. All these papers base their identification strategy of the average COVID-19 effect on the cross-country differences in the implementation of lockdown measures over time and study treatment effect heterogeneity by focusing on subsamples or interacting the treatment variable with other possible determinants of heterogeneity. We share with these studies the ambition to estimate the causal impact of COVID-19 on trade and its possible heterogeneity. However, we use a different approach based on constructing a counterfactual using the predictive power of ML that, as explained above, recognises that all firms are directly or indirectly affected by this economy-wide shock and that it is very challenging to define ex-ante a variable summarising the (differential) intensity of the shock for each firm. Moreover, we implement the heterogeneity analysis by using a classification analysis that safeguards against the risks of overfitting and multiple testing. Among the possible determinants of heterogeneity, we also consider a firm's diversification on the export and import side. Therefore, our study is also related to the international trade literature on the role of diversification in mediating the impact of adverse shocks (Kramarz et al., 2020; Grossman et al., 2021; Lafrogne-Joussier et al., 2022).

Using our innovative ML approach, we find that the COVID-19 shock reduced the probability of a Colombian firm surviving in the export market in April 2020 by around 20 percentage points. Our analysis of the estimated distribution of treatment effects shows that there is considerable heterogeneity behind these average results. We highlight that more affected firms tend to be small-sized and more exposed to export destinations and import source countries that are more severely hit by the containment policies related to the COVID-19 shock. We identify the firms most and least affected by COVID-19 and compare their characteristics by combining the Sorted Partial Effects methodology with our causal ML approach, which shows that integration into global value chains on the import side is an important determinant of exporters' resilience to the COVID-19 shock. Our findings contribute to the development of targeted recovery policies by identifying the firms most affected by exogenous widespread shocks.

The paper is structured as follows. Section 2 details our empirical strategy. Section 3 presents the firm-level data, variables employed in the analysis, and descriptive statistics. Section 4 reports the estimation results, and Section 5 summarizes our findings and discusses the relevance and limitations of our analysis.

2 Methodological framework

This section lays out our empirical approach to estimating the effect of an economy-wide shock on firms' survival probabilities in export markets and exploring its heterogeneity based on firms' observable attributes.

2.1 Our Causal ML setup

We aim to study the (conditional) average effect of an economy wide shock (e.g., the COVID-19 effect in our specific application) on the probability that the cohort of firms that were exporting in a given month during the pre-shock year t_{s-1} (e.g., year 2019 in our specific application) will export again during the same month of the year of the shock t_s (that is 2020 in our specific application).[4] Therefore, the empirical analysis is carried out separately for each month.[5] This allows the effects of the explanatory variables (e.g., the hypothesized determinants of firm export status) to vary throughout the year.

For economy-wide shocks such as COVID–19, there is no unambiguous definition of an "untreated" group because, plausibly, all firms are subject to the shock. Consequently, if we define the potential outcome for firm i at time t under treatment status $D \in \{0, 1\}$ as Y_{it}^D—where D indicates the presence of the shock—the standard Conditional Independence Assumption (CIA), $Y_{i,t_s}^0 \perp\!\!\!\perp D_{i,t_s} \mid X_{i,t_{s-1}}$, used to identify the Average Treatment Effect on the Treated (ATT) when a contemporaneous control group is available cannot be invoked as the assumption of common support is violated since $P(D_{i,t_s} = 1 \mid X_{i,t_{s-1}}) = 1$. Indeed, in this setting, the ATT coincides with the Average Treatment Effect (ATE) for the cohort of COVID–19-exposed firms, ATE $= \mathbf{E}(Y_{i,t_s}^1 - Y_{i,t_s}^0)$.[6] Furthermore, even an identification strategy based on comparing individual firms subject to different treatment intensities does not seem feasible due to the complex and ex-ante unknown paths through which firms are potentially exposed to treatment. Though we study whether treatment effect heterogeneity depends, inter alia, on firm-specific measures of the intensity of the COVID-19 shock,[7] the intensity of treatment effect might also depend on other firms' characteristics, such as the identity of suppliers and clients, the characteristics of the traded final product, among many

[4] Although the primary analysis focuses on the extensive margin, the proposed methodology is general and can be readily extended to continuous outcomes, allowing an analysis of the intensive margin. Descriptive evidence for the intensive margin is provided in the Appendix.

[5] In line with the literature on this topic (see, e.g., de Lucio et al., 2020; Berthou and Stumpner, 2024; Espitia et al., 2021; Liu et al., 2021), we perform a monthly analysis as the COVID-19 shock has evolved rapidly and unevenly over the months in 2020, as described in Online Appendix A.

[6] While we assume ATT = ATE because we think that all firms are subject to the economy-wide COVID-19 shock, this does not imply that all firms experience a non-zero effect. Some treated firms may have a negligible or even positive impact from the shock. In such cases, our ATT estimate is not biased but simply reflects treatment–effect heterogeneity. However, if some firms were not in fact exposed to the COVID-19 shock, our methodology would yield an ATT estimate that is biased toward zero.

[7] These indexes are described in detail in section 3. They are based on firms' past export and import activities in different countries and on the time-varying strength of the virus and the stringency of the policies aimed at mitigating its spread.

others, that we cannot know in advance and whose interactions are *a priori* unknown.

Therefore, we refer to all the observations $(Y_{i,t_s}, X_{i,t_{s-1}})$ for the cohort of firms that exported in t_{s-1} as the treated group (i.e., all observations belonging to our sample at t_s). Moreover, invoking the so-called consistency assumption, we assume that for the treated group the observed outcome in the year of the shock Y_{i,t_s} represents the potential outcome in case of treatment Y^1_{i,t_s}.[8]

As is common when studying the effects of widespread shocks, we must therefore use the information about behavior before the shock to estimate the counterfactual behavior (in the hypothetical situation without the shock) during the actual shock. This process involves forecasting the future conduct of entities based on their historical behavior, an application perfectly suited to ML techniques, which are designed for such out-of-sample prediction tasks. In line with the reasoning of Varian (2016), and drawing parallels with the applications employed by Cerqua and Letta (2020), and Fabra et al. (2022), we harness the predictive strength of ML techniques. This allows us to construct a hypothetical scenario for firm-level outcomes during the shock period, using pre-shock data concerning firms' export behaviors and attributes. We will use information on the export status in a given month of t_{s-1} (that is 2019 in our specific application) for firms that were exporting in the same month of year t_{s-2} (e.g., year 2018 in our specific application) and the observed characteristics of such firms in t_{s-2} to learn the counterfactual function that we apply to the treated group for estimating $Y^0_{it_s}$. We refer to the observations $(Y_{it_{s-1}}, X_{it_{s-2}})$ for the cohort of firms that exported in t_{s-2} as the control group.

The main assumptions used to reconstruct the unobserved counterfactual outcome during the year of the shock using the pre-shock observed firms' behaviour are: (i) absence of anticipatory effects of the shock on covariates measured at t_{s-1} and t_{s-2} and on the outcome at t_{s-1}, (ii) stability of the counterfactual function in time and (iii) a common support assumption. They are explained in detail below.

(i) *No anticipation effects on outcomes and covariates.* Neither observed outcomes at $t_s - 1$ nor covariates at $t_s - 2$ and $t_s - 1$ are affected by the shock happening at t_s:

$$Y_{i,t_{s-1}} = Y^0_{i,t_{s-1}}, \quad X_{i,t} = X^0_{i,t} \quad \text{for} \quad t = (t_s - 1, t_s - 2) \tag{1}$$

[8]The consistency assumption corresponds to the first component of the Stable Unit Treatment Value Assumption (SUTVA; Keele, 2015), which requires that there are no hidden forms of treatment. We maintain this assumption by adopting a deliberately broad definition of treatment — being in the sample at time t_s — which, while encompassing a potentially different intensity of the shock for each unit, remains useful for generating policy recommendations concerning the firms relatively more affected by the shock. These recommendations are informative even if treatment-effect heterogeneity is partly confounded by heterogeneity in the treatment itself, since firms that are more affected may be so either because their characteristics are correlated with higher treatment intensity or because the treatment interacts with their characteristics. The second component of SUTVA — the no-interference assumption — is not relevant in our setting because we do not rely on a contemporaneous control group that could be indirectly affected and our focus is on estimating the total effect of the treatment, which by construction includes both the direct effect on a unit from the treatment it receives and any indirect effects arising from spillovers or general-equilibrium adjustments. Disentangling between direct and indirect effects lies beyond the scope of this paper.

Notice that, since the treatment occurs at t_s, for (1) to hold it is sufficient to rule out any effect of the future shock at t_s on the observed $Y_{i,t_{s-1}}$. Moreover, (1) implies that the CIA holds for the control group at t_{s-1}, that is, $Y^0_{i,t_{s-1}} \perp\!\!\!\perp D_{i,t_{s-1}} \mid X_{i,t_{s-2}}$. Consequently, at t_{s-1} the conditional expectation of the observed outcome coincides with that of the potential outcome under no treatment: $\mathbf{E}[Y^0_{i,t_{s-1}} \mid X_{i,t_{s-2}}] = \mathbf{E}[Y_{i,t_{s-1}} \mid X_{i,t_{s-2}}]$.

(ii) *Stability of the Counterfactual Function.* This assumption is about the stability of the function that expresses the expected value of the conditional potential outcome in time. Define $Y^0_{i,t} = f^0_t(X^0_{i,t-1}) + u^0_{i,t}$, where $f^0_t(\cdot)$ is a generic model or function representing the relationship between explanatory variables and the outcome in the absence of the shock such that $\mathbf{E}[Y^0_{i,t}|X^0_{i,t-1}] = f^0_t(X^0_{i,t-1})$. Under (i), for $t = t_s - 1$ we have that $Y_{i,t_s-1} = f^0_{t_s-1}(X_{i,t_s-2}) + u^0_{i,t_s-1}$ such that $\mathbf{E}[Y_{i,t_s-1}|X_{i,t_s-2}] = f^0_{t_s-1}(X_{i,t_s-2})$. The second assumption states that the function f^0_t does not depend on t, i.e., it is stable over the two considered years:

$$f^0_{t_s-1} = f^0_{t_s} = f^0. \tag{2}$$

Under assumptions (i) and (ii), if $X_{i,t_{s-1}} = X_{i,t_{s-2}}$ then $\mathbf{E}[Y^0_{i,t_s} \mid X_{t_{i,s-1}}] = \mathbf{E}[Y^0_{i,t_{s-1}} \mid X_{i,t_{s-2}}]$. In other words, the conditional expectation of the potential outcome under no treatment at t_s coincides with that at t_{s-1}.

(iii) *Common support.* The support of the distribution of the explanatory variables of the firms belonging to the treated group is included in the support of the distribution of the explanatory variables of the firms belonging to the control group:

$$P(D_{i,\{t_s,t_s-1\}} = 1|X_{i,\{t_s-1,t_s-2\}}) = e(X_{i,\{t_s-1,t_s-2\}}) < 1 \tag{3}$$

where $D_{i,\{t_s,t_s-1\}}$ is a dummy variable indicating whether an observation belongs to the treated group or to the control group, and $X_{i,\{t_s-1,t_s-2\}}$ are the corresponding explanatory variables. Therefore, this expression defines a condition on the values of the propensity score, which we denote as $e(X_{i,\{t_s-1,t_s-2\}})$. This assumption allows nonparametric identification of the (conditional) average effects.

Thanks to these assumptions, when $X_{i,t_{s-1}} = X_{i,t_{s-2}}$, we have that $\mathbf{E}[Y^0_{i,t_s} \mid X_{i,t_{s-1}}] = \mathbf{E}[Y^0_{i,t_{s-1}} \mid X_{i,t_{s-2}}] = \mathbf{E}[Y_{i,t_{s-1}} \mid X_{i,t_{s-2}}]$: the conditional expectation function of the potential outcome in case of no treatment at t_{s-1} can be identified by computing the conditional expectation function of the observed outcome at t_{s-1} and it coincides with the conditional expectation function of the potential outcome in case of no treatment at t_s.

Since f^0 is in practice unknown, we must estimate it. Under the above assumptions, we can write $Y^0_{i,t_s} = f^0(X_{i,t_s-1}) + u^0_{i,t_s}$, such that $\mathbf{E}[Y^0_{i,t_s}|X_{i,t_s-1}] = f^0(X_{i,t_s-1})$, and we can use data on $t_s - 2$ and $t_s - 1$ to estimate $Y^0_{i,t_s-1} = f^0(X_{i,t_s-2}) + u^0_{t_s-1}$ and retrieve $\hat{f}^0$. By applying this invariant estimated function to the covariates of $t_s - 1$, we can obtain the

predictions for the counterfactual (without the shock) outcome in t_s:

$$\hat{Y}^0_{i,t_s} = \hat{f}^0(X_{i,t_s-1}) = Y^0_{i,t_s} - \overbrace{\mathcal{E}^0_{i,t_s}}^{Prediction\ error} - \overbrace{u^0_{i,t_s}}^{Orthogonal\ error}. \tag{4}$$

The model we utilize to derive this counterfactual (and the counterfactual itself) is referred to as the "Shock Unaware Machine" (SUM), a term acknowledging the ML techniques employed in constructing the counterfactual and the fact that no information about the shock is used in the analyses. In the application we present in this paper, we rely on the "K-fold" cross-validation method (with $K = 5$) to discriminate between the considered ML techniques. We randomly divide the set of exporters observed in $t_s - 2 = 2018$ (considering the exporting success during the same month in $t_s - 1 = 2019$ as the outcome) into 5 equally sized groups and obtain the predictions for the firms belonging to a group by estimating $Y_{i,2019} = f^0(X_{i,2018}) + u^0_{i,2019}$ with different ML models on the firms belonging to the other groups. Then we compute the accuracy of the different models for each month and choose the model with the best average performance across months. Notice that this comparison is entirely based on the pre-pandemic accuracy of the ML models by comparing the predictions $\hat{Y}_{i,2019}$ with the observed $Y_{i,2019}$, not on its merits in predicting the firms' outcomes in 2020. Finally, we obtain the $\hat{Y}^0_{i,2020}$ by estimating $Y_{i,2019} = f^0(X_{i,2018}) + u^0_{i,2019}$ on the entire set of 2018 units (also in this case month by month) and, as shown in (4), applying the estimated function $\hat{f}^0$ to the set of 2019 units. Given that during the first three months of 2020 Colombia was in practice not exposed to COVID-19 (and therefore $Y_{i,2020} = Y^0_{i,2020}$), if assumption (2) holds, we expect that in those months the accuracy of the predictions $\hat{Y}_{i,2019}$ obtained in the cross-validation step for 2019 will be very similar to the accuracy of $\hat{Y}^0_{i,2020}$ for 2020.

Following Cerqua and Letta (2020) and Fabra et al. (2022), we define the simple comparison of the observed outcome under the shock in t_s with the estimated counterfactual outcome as an estimator of the individual-specific shock effect α_i. This comparison is represented as:

$$\hat{\hat{\alpha}}_i = Y_{i,t_s} - \hat{Y}^0_{i,t_s}. \tag{5}$$

This provides the full distribution of estimated individual treatment effects, that is, each unit's Conditional Average Treatment Estimate (CATE) estimate (Salditt et al., 2024).

The ATE and the CATE$_z$, that is the Conditional Average Treatment Effect for those units with $Z_{i,t_s-1} = z_{i,t_s-1}$ where Z is a subset of the variables X ($Z \subset X$),[9] are estimated by averaging these estimated individual treatment effects. Therefore, the estimators of ATE

[9]For example, CATE$_{textile}$ is the average treatment effect for firms belonging to the textile industry.

and CATE_z based on $\hat{\hat{\alpha}}_i$ can be defined as:

$$\bar{\hat{\hat{\alpha}}} = \frac{1}{N}\sum_{i=1}^{N}\hat{\hat{\alpha}}_i, \qquad \bar{\hat{\hat{\alpha}}}_Z = \frac{1}{N_z}\sum_{i\in C_z}\hat{\hat{\alpha}}_i, \tag{6}$$

where N is the number of observations, $C_z = \{i : Z_{i,t_{s-1}} = z_{i,t_{s-1}}\}$ and $N_z = |C_z|$.

As it is shown below, $\bar{\hat{\hat{\alpha}}}_z$ is an unbiased estimator of CATE_z if in the relevant subsample the mean of the expected prediction error, $(1/N_z)\sum_{i\in C_z}\mathbf{E}[\mathcal{E}^0_{i,t_s}]$, is zero.

$$\begin{aligned}
\mathbf{E}\Big[\bar{\hat{\hat{\alpha}}}_z\Big] &= \mathbf{E}\left[\frac{1}{N_z}\sum_{i\in C_z}\hat{\hat{\alpha}}_i\right] = \frac{1}{N_z}\sum_{i\in C_z}\mathbf{E}[\hat{\hat{\alpha}}_i] \\
&= \frac{1}{N_z}\sum_{i\in C_z}\mathbf{E}\big[Y_{i,t_s} - Y^0_{i,t_s} - \mathcal{E}^0_{i,t_s} - u^0_{i,t_s}\big] \\
&= \underbrace{\frac{1}{N_z}\sum_{i\in C_z}\alpha_i}_{CATE_z} - \frac{1}{N_z}\sum_{i\in C_z}\mathbf{E}[\mathcal{E}^0_{i,t_s}] - \frac{1}{N_z}\sum_{i\in C_z}\underbrace{\mathbf{E}[u^0_{i,t_s}]}_{=0 \text{ by (i) and (ii)}}
\end{aligned} \tag{7}$$

Notice that in the second row, we substitute Eq. (4) plugged in Eq. (5). Similarly, $\bar{\hat{\hat{\alpha}}}$ is an unbiased estimator of ATE if the mean of the expected prediction error, $(1/N)\sum_{i=1}^{N}\mathbf{E}[\mathcal{E}^0_{i,t_s}]$, is zero, and $\bar{\hat{\hat{\alpha}}}_i$ is an unbiased estimator of the unit i's CATE if $\mathbf{E}[\mathcal{E}^0_{i,t_s}] = 0$.

In what follows, we introduce alternative estimators that compare the predicted counterfactual outcomes with predicted outcomes under the treatment scenario, rather than directly with the observed treatment outcomes. The first step is to decompose the observed outcome in t_s under the shock, Y^1_{i,t_s}, into a generic model or function $f^1(X^1_{i,t_s-1})$, capturing the relationship between covariates and the outcome during the shock, and a residual component u^1_{i,t_s} orthogonal to the covariates:

$$Y^1_{i,t_s} = f^1(X^1_{i,t_s-1}) + u^1_{i,t_s}, \quad s.t.\ \mathbf{E}[Y^1_{i,t_s} \mid X^1_{i,t_s-1}] = f^1(X^1_{i,t_s-1}). \tag{8}$$

Given that $Y^1_{i,t_s} = Y_{i,t_s}$ and $X^1_{i,t_s-1} = X_{i,t_s-1}$, we can write:

$$Y_{i,t_s} = f^1(X_{i,t_s-1}) + u^1_{i,t_s}, \quad s.t.\ \mathbf{E}[Y_{i,t_s} \mid X_{t_s-1}] = f^1(X_{i,t_s-1}). \tag{9}$$

We then define an alternative estimator of the individual-specific shock effect α_i, that is, each unit's Conditional Average Treatment Estimate (CATE) estimate (Salditt et al., 2024), as the difference between the predicted outcome under the shock in t_s and the predicted counterfactual outcome in the absence of the shock for the same firm:

$$\hat{\alpha}_i = \hat{Y}^1_{i,t_s} - \hat{Y}^0_{i,t_s}, \tag{10}$$

where $\hat{Y}^1_{i,t_s} = \hat{f}^1(X_{i,t_s-1}) = Y_{i,t_s} - \mathcal{E}^1_{i,t_s} - u^1_{i,t_s}$.

We refer to the model used to predict Y_{i,t_s} (and the predictions $\hat{Y}_{i,t_s}$ themselves) as the *Shock Aware Machine* (SAM). The term "Shock Aware" emphasizes that this model exploits information from the observed shock scenario. Importantly, SAM predictions are expressed in the same metric as the counterfactual predictions, which are produced by the SUM, allowing them to be directly comparable.[10] In our application, the SAM expresses the outcome in 2020 of exporters operating the foreign market in 2019 as a function of their characteristics in 2019 and the information about governments' shock-related stringency measures all over the world coming from Hale et al. (2020).[11] Similarly to the procedure followed to select the best-performing SUM, we rely on a 5-fold cross-validation strategy to obtain a 2020 prediction for each firm that exported in 2019. We randomly group the 2019 exporters into five equally sized subsets and we predict the 2020 outcomes of the firms contained in one subset by using the information of firms contained in the remaining four subsets. In other words, we train the models on a random 80% of the data and test them on the remaining 20% and we repeat the process five times for each different 20% subset, thus obtaining a 2020 prediction for each 2019 exporter.

The ATE estimator and CATE_z estimator based on $\hat{\alpha}_i$ are:

$$\bar{\hat{\alpha}} = \frac{1}{N}\sum_{i=1}^{N}\hat{\alpha}_i, \qquad \bar{\hat{\alpha}}_z = \frac{1}{N_z}\sum_{i\in C_z}\hat{\alpha}_i, \tag{11}$$

As shown below, $\bar{\hat{\alpha}}_z$ is an unbiased estimator of the CATE_z if, in the relevant subsample, the mean of the expected prediction-error difference between SAM and SUM, $\frac{1}{N_z}\sum_{i\in C_z}\mathbf{E}[\mathcal{E}^1_{i,t_s} - \mathcal{E}^0_{i,t_s}]$, is zero:

$$\begin{aligned}
\mathbf{E}\big[\bar{\hat{\alpha}}_z\big] &= \frac{1}{N_z}\sum_{i\in C_z}\mathbf{E}[\hat{\alpha}_i] \\
&= \frac{1}{N_z}\sum_{i\in C_z}\mathbf{E}\big[Y_{i,t_s} - Y^0_{i,t_s} - \big(\mathcal{E}^1_{i,t_s} - \mathcal{E}^0_{i,t_s}\big) - \big(u^1_{i,t_s} + u^0_{i,t_s}\big)\big] \\
&= \underbrace{\frac{1}{N_z}\sum_{i\in C_z}\alpha_i}_{CATE_z} - \frac{1}{N_z}\sum_{i\in C_z}\mathbf{E}[\mathcal{E}^1_{i,t_s} - \mathcal{E}^0_{i,t_s}] - \underbrace{\frac{1}{N_z}\sum_{i\in C_z}\mathbf{E}[u^1_{i,t_s} - u^0_{i,t_s}]}_{=0 \text{ by (i), (ii) and (8)}}.
\end{aligned} \tag{12}$$

Similarly, $\bar{\hat{\alpha}}$ is an unbiased estimator of the ATE if $\frac{1}{N}\sum_{i=1}^{N}\mathbf{E}[\mathcal{E}^1_{i,t_s} - \mathcal{E}^0_{i,t_s}] = 0$, and $\hat{\alpha}_i$ is an unbiased estimator of the unit i's CATE if $\mathbf{E}[\mathcal{E}^1_{i,t_s} - \mathcal{E}^0_{i,t_s}] = 0$.

Given the definitions of SAM and SUM, to simplify the reasoning in the following, we

[10] Notice that with $\hat{\alpha}_i$, we are comparing a probability (counterfactual) with a binary value (observed outcome), while with $\hat{\alpha}_i$, we are comparing two estimated probabilities.

[11] See subsection 3.1. We do not introduce these variables explicitly as an argument of $f^1()$ to simplify notation.

refer to Eqs. (5) and (10), respectively as

$$\hat{\hat{\alpha}}_i = Y - \hat{Y}_{SUM} = Y - SUM. \tag{13}$$

$$\hat{\alpha}_i = \hat{Y}_{SAM} - \hat{Y}_{SUM} = SAM - SUM. \tag{14}$$

The conditions behind these identification results are not directly testable as they are expressed in terms of the expected values of the prediction error $\mathcal{E}^0_{i,t_s}$ that is a function of the unobservable counterfactual Y^0_{i,t_s}. Table 1 distinguishes the five different scenarios concerning the expected values of $\mathcal{E}^0_{i,t_s}$ and $\mathcal{E}^1_{i,t_s}$ that are relevant in determining whether applying the statistic $\mathbf{T}$ to $Y - SUM$ and $SAM - SUM$ is able to recover the corresponding treatment effect estimand (e.g., whether averaging the estimated individual treatment effects would recover the average treatment effect).

Table 1: Identification of generic functions of the individual treatment effects, $\mathbf{T}$, according to the corresponding value taken by the prediction errors

	$\mathbf{T}(SAM - SUM)$	$\mathbf{T}(Y - SUM)$
$\mathbf{T}[\mathbf{E}[\mathcal{E}^1_{i,t_s}]] \neq 0$ and $\mathbf{T}[\mathbf{E}[\mathcal{E}^0_{i,t_s}]] = 0$	×	✓
$\mathbf{T}[\mathbf{E}[\mathcal{E}^1_{i,t_s}]] = \mathbf{T}[\mathbf{E}[\mathcal{E}^0_{i,t_s}]] = 0$	✓	✓
$\mathbf{T}[\mathbf{E}[\mathcal{E}^1_{i,t_s}]] = 0$ and $\mathbf{T}[\mathbf{E}[\mathcal{E}^0_{i,t_s}]] \neq 0$	×	×
$\mathbf{T}[\mathbf{E}[\mathcal{E}^1_{i,t_s}]] = \mathbf{T}[\mathbf{E}[\mathcal{E}^0_{i,t_s}]] \neq 0$	✓	×
$\mathbf{T}[\mathbf{E}[\mathcal{E}^1_{i,t_s}]] \neq \mathbf{T}[\mathbf{E}[\mathcal{E}^0_{i,t_s}]] \neq 0$	×	×

The estimators based on Y-SUM identify the population parameters when $\mathbf{T}[\mathbf{E}[\mathcal{E}^0_{i,2020}]] = 0$. The estimators based on SAM-SUM are unbiased whenever $\mathbf{T}[\mathbf{E}[\mathcal{E}^1_{i,2020}]] = \mathbf{T}[\mathbf{E}[\mathcal{E}^0_{i,2020}]]$. Under the assumption that the strength of the COVID-19 effect on export propensity was at most very limited during the first quarter of 2020, we will use the out-of-sample prediction errors for the first quarter of 2020 as a proxy for the unobservable behavior of $\mathcal{E}^0_{i,2020}$ in the following months. Moreover, as explained in detail in section 4.2, the distribution of the estimated treatment effects during the first quarter will be used to check the credibility of the above assumptions for the set of all 2019 exporters and for different subsets of 2019 exporters defined according to their characteristics $X_{i,2019}$ or to their position in the distribution of such effects.

The inference is performed using bootstrap. Specifically, bootstrap samples are drawn by resampling the training and testing datasets with replacement. To construct confidence intervals, the predictions across all bootstrap replicates are aggregated, and the empirical distribution of the resulting estimates is used to calculate the percentile-based bounds for the 95% confidence interval, thus capturing the uncertainty in the predicted effects due to sampling variability. However, as a note of caution, we remark that the literature on

causal ML has shown that estimators based on ML estimation of the conditional expectation function of potential outcomes, such as $\bar{\hat{\alpha}}$ and $\bar{\hat{\alpha}}$, inherit the slow convergence rates of the ML method on which they are based and are not asymptotically normal, making inference problematic. The problem with these estimators is that the moment conditions on which they are based are not Neyman-orthogonal. To do a robustness check for the average treatment effects, we use the AIPW-Double ML estimator (with 5-fold cross-fitting and nuisance parameters estimated with Random Forest) that is consistent and asymptotically normal for $ATT = ATE = \mathbf{E}(Y^1_{t_s} - Y^0_{t_s})$ and $ATE_{\{t_s,t_s-1\}} = \mathbf{E}(Y^1_{\{t_s,t_s-1\}} - Y^0_{\{t_s-1,t_s\}})$ (Chernozhukov et al., 2018), in which the average potential outcomes are expressed with a moment condition which is Neyman Orthogonal. $ATT = ATE = \mathbf{E}(Y^1_{t_s} - Y^0_{t_s})$ is the parameter we aim to estimate with $\bar{\hat{\alpha}}$ and $\bar{\hat{\alpha}}$. $ATE_{\{t_s,t_s-1\}} = \mathbf{E}(Y^1_{\{t_s,t_s-1\}} - Y^0_{\{t_s,t_s-1\}})$ is the average treatment effect obtained considering the cohorts of treated and control firms together as a unique sample. To identify $ATE_{\{t_s,t_s-1\}}$, we have to make the following additional assumptions: $Y^1_{t_s} \perp\!\!\!\perp D_{t_s} \mid X_{t_{s-1}}$; $\mathbf{E}[Y^1_{t_s} \mid X_{t_{s-1}}] = \mathbf{E}[Y^1_{t_{s-1}} \mid X_{t_{s-2}}]$; $Y_{t_s} = Y^1_{t_s}$; $P(D_{\{t_s,t_s-1,\}} = 1 | X_{\{t_s-1,t_s-2\}}) = e(X_{\{t_s-1,t_s-2\}}) > 0$. Indeed, these assumptions are needed to identify the Average Treatment Effect on the Untreated (ATU), which is defined as $ATU = ATE_{t_{s-1}} = \mathbf{E}(Y^1_{t_{s-1}} - Y^0_{t_{s-1}})$, because the $ATE_{\{t_s,t_s-1\}} = ATU * (1 - \sigma(\cdot)) + ATT * (\sigma(\cdot))$ where $\sigma(\cdot)$ represents the share of the treated population. Let's define the following pseudo-outcome[12]

$$\tilde{Y}^{ATE}_{\{t_s,t_s-1\}} = \underbrace{\hat{Y}^1_{\{t_s,t_s-1\}} - \hat{Y}^0_{\{t_s,t_s-1\}}}_{\text{outcome predictions}} + \underbrace{\frac{D_{\{t_s,t_s-1\}}(Y_{\{t_s,t_s-1\}} - \hat{Y}^1_{\{t_s,t_s-1\}})}{\hat{e}(X_{\{t_s-1,t_s-2\}})} - \frac{(1 - D_{\{t_s,t_s-1\}})(Y_{\{t_s,t_s-1\}} - \hat{Y}^0_{\{t_s,t_s-1\}})}{1 - \hat{e}(X_{\{t_s-1,t_s-2\}})}}_{\text{weighted residuals}} \tag{15}$$

The AIPW-Double ML estimator (where $\hat{Y}^1$, $\hat{Y}^0$ and $\hat{e}$ are estimates of the nuisance parameters obtained with ML by cross-fitting) is the average of these pseudo-outcomes and identifies the $ATE_{\{t_s,t_s-1\}}$.[13]

Finally, following the nomenclature introduced by Künzel et al. (2019), we note that our estimator $\bar{\hat{\alpha}}_i$ is referred to in the literature as the *T-learner*. In particular, we are applying a T-Learner to the group of treated units. Although the T-learner is a consistent estimator for CATE, it may have shortcomings in finite samples due to a phenomenon known as *regularization bias*. This issue arises because the outcome models for treated and control units, denoted $\hat{f}_1(x)$ and $\hat{f}_0(x)$, are estimated separately and may be subject to different degrees of regularization. Such discrepancies are particularly pronounced when the sample

[12] For notational convenience, the subscript "i" is omitted in the following expression.

[13] The procedure consists of the following steps: (a) randomly partition the data into K equally sized folds; (b) for each fold k, leave it out and use the remaining $K-1$ folds to estimate the nuisance functions $\mu(d,x) = \mathbf{E}[Y_i \mid D_i = d, X_i = x]$ and $e_d(x)$; (c) predict the nuisance parameters on the left-out fold k using the estimated models, yielding cross-fitted values $\hat{\mu}^{-k}(d,x)$ and $\hat{e}_d^{-k}(x)$; and (d) repeat steps (a)-(c) so that each fold serves once as the validation set (Knaus, 2022). This cross-fitting approach ensures that no observation is used to predict its own nuisance parameters, thereby reducing overfitting and guaranteeing the validity of inference. These cross-fitted estimates are subsequently used to compute the pseudo-outcomes.

sizes for the two groups differ substantially, and the machine learning methods used employ regularization schemes that are sensitive to sample size. In this case, the model trained on the smaller group may be oversmoothed, potentially introducing artificial variation into the estimated treatment effect.

A second concern relates to *regularization-induced confounding*, which may occur when the covariate distributions of treated and control units are not well aligned. Under such circumstances, the models $\hat{f}^1(x)$ and $\hat{f}^0(x)$ may be effectively trained on disjoint regions of the covariate space. Consequently, their difference could reflect underlying distributional shifts rather than true treatment effect heterogeneity. This issue is often associated with settings where the estimated outcome models correspond to regions with substantially different values of the propensity score—for example, where $\hat{f}_1(x)$ is learned predominantly in areas with high propensity scores and $\hat{f}_0(x)$ in areas with low propensity scores. Since the T-learner does not explicitly adjust for the propensity score or reweight observations to balance treatment groups, it may be more susceptible to this type of bias compared to alternative estimators considering also the distribution of the propensity score, such as the DR-learner introduced in Kennedy et al. (2020) (that approximate the $\mathbf{E}[\tilde{Y}^{ATE}_{\{t_s-1,t_s\}}|X]$ as a generic ML problem; the reader is referred to the Appendix for further details).

However, in our application, these concerns are likely to be minimal. First, we leverage a large sample size, which mitigates the risk of over-regularization. Moreover, we employ a range of ML methods with varying levels of regularization intensity and obtain highly consistent results across estimators. Second, the distribution of covariates is balanced between treated and control units as shown by the estimated propensity score which is nearly identical across groups (see Figure Appx.3). This implies that both $\hat{f}_1(x)$ and $\hat{f}_0(x)$ are estimated over comparable regions of the covariate space, thereby limiting the potential for regularization-induced confounding. Third, in practice (see Table 6, Table 7, Table 8 and the monthly comparison of the distribution of estimated CATE in the Online Appendix), the CATE results obtained for the T-Learner are very similar to those obtained with the other meta-learners using the propensity score and with Causal Forest within the Generalized Random Forests framework.

2.2 Treatment effect heterogeneity analysis

As a further step, we perform the heterogeneity analysis by adapting the Sorted Partial Effect (SPE) method introduced in Chernozhukov et al. (2018). Formally, the SPEs are defined as percentiles of the α_i, the individual Treatment Effects (TE), and can supply a more detailed summary of the distribution of TE than the Average Treatment Effects (ATE), commonly employed in econometric analysis. The SPEs are defined as

$$\alpha^*(u) = u^{th} - \text{ percentile of } \alpha_i. \tag{16}$$

In our setting, $\alpha^*(u)$ is a function of X_{t_s-1} defined over its distribution in the population of $t_s - 1$ exporters.

The SPEs are used to do a classification analysis (CA) that allocates the $t_s - 1$ exporters into two groups, the most and the least affected by the shock, according to whether their α are lower than $\alpha^*(25)$ or greater than $\alpha^*(75)$, respectively. Notice that, since the shock effect is negative, we have defined as the most (negative) affected units those whose α lie in the left tail of the sorted distribution of treatment effects. Finally, to study the determinants of treatment effect heterogeneity, we focus on the difference in means ($CADiff$) of the X_{t_s-1} across the most and least affected groups. In the estimation, we use sample analogues of $\alpha^*(u)$ and $CADiff$. We calculate standard errors of $\alpha^*(u)$ and $CADiff$ by bootstrapping the entire estimation process, starting from the initial α_i estimation step.

Starting from B bootstrap replications of all the estimation steps (including the prediction stage), we calculate the $CADiff$ B times. To determine the significance of the $CADiff$ we perform a two-tailed test. The p-values are constructed as follows:

$$2 \cdot min\{Pr(S \geq t|H_0), Pr(S \leq t|H_0)\}$$

being t the observed t test statistic, $t = \frac{\widetilde{CADiff_{original}}}{\tilde{\sigma}}$, drawn from the unknown distribution S. $\tilde{\sigma}$ represents the standard deviation of the bootstrapped $CADiff$. To adjust the p-values and obtain the joint p-values taking into account that we are testing hypotheses jointly on many covariates, we reproduce the "single-step" method employed by Chernozhukov et al. (2018) to control for the family-wise error rate.[14]

[14]In the following is described the single-step algorithm. We will indicate the bootstrap version of a variable, v, as $\tilde{v}$ and its estimated version (on the original data) as $\hat{v}$. Moreover, $\Lambda(x)^{-u}$ will denote the first moment for the feature x of interest in the least affected group including the observational units i such that $\alpha_i < \alpha^*(u)$. Similarly, $\Lambda(x)^{+u}$ defines the first moment for the variable x of interest in the most affected group including the observational units i such that $\alpha_i > \alpha^*(1-u)$. Since we do not observe α directly, the mentioned quantities are estimated. According to the above convention, the estimated value of $\Lambda(x)^{-u}$ ($\Lambda(x)^{+u}$) will be $\hat{\Lambda}(x)^{-u}$ ($\hat{\Lambda}(x)^{+u}$) indicating the first moment for the variable x of interest for firms i such that $\hat{\alpha}_i > \hat{\alpha}^*(u)$ ($\hat{\alpha}_i < \hat{\alpha}^*(1-u)$). In the present paper $u = 25$, however, we will maintain the more general u notation for the sake of consistency with Section 4.
The single-step algorithm proceeds as follows: 1) for each variable $x \in X_t$, compute $\tilde{\Lambda}(x)^{+u}$ and $\tilde{\Lambda}(x)^{-u}$, bootstrap draws of $\hat{\Lambda}(x)^{+u}$ and $\hat{\Lambda}(x)^{-u}$ respectively. We want to test the null hypothesis, H_0, that $\Lambda^u(x) = 0$, for $\Lambda^u(x) = [\Lambda(x)^{-u}, \Lambda(x)^{+u}]$. 2) Construct a bootstrap draw of the distribution of $(\hat{\Lambda}^{+u}(x) - \hat{\Lambda}^{-u}(x))$, $Z^u_\infty(x)$. The latter is obtained by exploiting the bootstrap version of $\Lambda^{+u}(x)$ and $\Lambda^{-u}(x)$, namely: $\tilde{Z}_\infty(x) = \sqrt{n}(\tilde{\Lambda}^u(x) - \hat{\Lambda}^u(x))$ where $\tilde{\Lambda}^u(x) = [\tilde{\Lambda}(x)^{-u}, \tilde{\Lambda}(x)^{+u}]$ (similarly, $\hat{\Lambda}^u(x) = [\hat{\Lambda}(x)^{-u}, \hat{\Lambda}(x)^{+u}]$). 3) Repeat steps 1) and 2) B times; 4) compute a bootstrap estimator of the variance of Z_∞ as $\hat{\Sigma}^u(x) = \frac{q^u_{0.75}(x) - q^u_{0.25}(x)}{z_{0.75} - z_{0.25}}$ being $q^u_p(x)$ the pth sample quantile of $\tilde{Z}_\infty(x)$ and z_p the pth quantile of a standard normal distribution. 5) Use the latter to construct the test statistic $\tilde{\tau}(X_t) = sup_{x \in X_t} |\tilde{Z}_\infty(x)| \cdot |\hat{\Sigma}^u(x)|^{-1/2}$. A p-value for the null H_0 that $\Lambda^u(x) = 0$ for all $x \in X_t$ of the realization of the estimated statistic, $sup_{x \in X_t} |\hat{\Lambda}^u(x)| \cdot |\hat{\Sigma}(x)|^{-1/2} = s$, is given by the average number of times that $\tilde{\tau}(X_t)$ is greater than s, where $s = \frac{\beta^m_{1,f}}{\hat{\Sigma}^u(x)}$. The $\tilde{}$ indicates simply that the $\beta^m_{1,f}$ has been projected to the bootstrap dimension. Technically, the validity of the single-step correction relies on the fact that the sorted effects and their induced classification structure are smooth functionals of the data-generating process. In particular, Chernozhukov et al. (2018) show that the sorting and classification operators are Hadamard differentiable, which allows the use of the functional delta method to derive the large-sample distribution of the estimated effects and their differences.

The application of the SPE technique presents several advantages in our setting. First, the estimated $\alpha^*(u)s$ provide a summary of the distribution of the estimated treatment effects and, therefore, of treatment effect heterogeneity. Second, the CA identifies the subgroup of the population that is more affected by the treatment and the $CADiff$ studies how the heterogeneity of the treatment effect depends on observables without imposing (additional) functional form assumptions. Third, the $CADiff$ step provides p-value adjustments to account for the joint testing of all the covariates that are considered to detect if observables are associated with treatment effect heterogeneity.

2.3 Comparison with Generic ML

Our approach in estimating the individual treatment effect and in performing the heterogeneity analysis is similar to the generic ML technique presented in Chernozhukov et al. (2023), which is adapted to a situation in which there is no available (contemporaneous) control group (i.e., it is difficult to identify ex-ante firms that are not affected by the shock).

We show that our empirical strategy is built on the same pillars as Chernozhukov et al. (2023), but applies them to a different setting. To simplify the exposition, we refer to Table 2 which provides a simplified representation of our empirical setting.

Chernozhukov et al. (2023) deal with an experimental empirical setting in which one can easily separate a treated group from a control group. In order to study the heterogeneity of the average treatment effect, the first step of Chernozhukov et al. (2023) is to split randomly the sample under analysis in an auxiliary (A) and a main sample (M) of approximately the same size. Then, they employ ML techniques to learn in A the function approximating the potential outcomes in the treatment and non-treatment scenarios, while M is used to make inferences on the key features of treatment effect heterogeneity. In other words, they estimate the function describing the outcome in case of treatment (no treatment) on the subset of treated (non-treated) firms contained in A. These two estimated functions are used to impute the two potential outcomes for each firm contained in the M sample (the difference represents the estimated individual treatment effects) and study the treatment effect heterogeneity estimated for these firms by using, inter alia, the Sorted Effects method (Chernozhukov et al., 2018). This procedure is designed in this way to avoid overfitting (i.e., doing learning and prediction using the same sample), and, starting from the random splitting, it is repeated many times in order to obtain many distributions of estimated treatment effects to which the Sorted Effects method is applied.

As an example, in Table 2, we represent 20 exporting firms observed in 2018 or in 2019. In the context of our setting, the strategy of Chernozhukov et al. (2018) would imply that the 2019 sample, for which we are interested in estimating the average treatment effect, should be divided in two, as shown in the last column of Table 2. However, in the COVID-19 scenario, one cannot easily separate treated and control units because COVID-19 imposes a (at least indirect) treatment over all units, hence preventing the possibility of discerning

between treated and controls.[15] Moreover, with respect to Chernozhukov et al. (2023), in our empirical setting, we do not have the necessity to predict the outcome of controls in the case of "no treatment" because we are not interested in estimating the COVID-19 effect on 2018 exporters.

Our Setting			Chernozhukov et al. (2023) Setting	
SUM		SAM	$A-M$ splitting	
$(X_{2018}, Y_{2019})_1$	$(X_{2019}, Y_{2020})_{11}$	$(X_{2019}, Y_{2020})_{11}$	$(X_{2019}, Y_{2020})_{11}$	
$(X_{2018}, Y_{2019})_2$	$(X_{2019}, Y_{2020})_{12}$	$(X_{2019}, Y_{2020})_{12}$	$(X_{2019}, Y_{2020})_{12}$	
$(X_{2018}, Y_{2019})_3$	$(X_{2019}, Y_{2020})_{13}$	$(X_{2019}, Y_{2020})_{13}$	$(X_{2019}, Y_{2020})_{13}$	A
$(X_{2018}, Y_{2019})_4$	$(X_{2019}, Y_{2020})_{14}$	$(X_{2019}, Y_{2020})_{14}$	$(X_{2019}, Y_{2020})_{14}$	
$(X_{2018}, Y_{2019})_5$	$(X_{2019}, Y_{2020})_{15}$	$(X_{2019}, Y_{2020})_{15}$	$(X_{2019}, Y_{2020})_{15}$	
$(X_{2018}, Y_{2019})_6$	$(X_{2019}, Y_{2020})_{16}$	$(X_{2019}, Y_{2020})_{16}$	$(X_{2019}, Y_{2020})_{16}$	
$(X_{2018}, Y_{2019})_7$	$(X_{2019}, Y_{2020})_{17}$	$(X_{2019}, Y_{2020})_{17}$	$(X_{2019}, Y_{2020})_{17}$	
$(X_{2018}, Y_{2019})_8$	$(X_{2019}, Y_{2020})_{18}$	$(X_{2019}, Y_{2020})_{18}$	$(X_{2019}, Y_{2020})_{18}$	M
$(X_{2018}, Y_{2019})_9$	$(X_{2019}, Y_{2020})_{19}$	$(X_{2019}, Y_{2020})_{19}$	$(X_{2019}, Y_{2020})_{19}$	
$(X_{2018}, Y_{2019})_{10}$	$(X_{2019}, Y_{2020})_{20}$	$(X_{2019}, Y_{2020})_{20}$	$(X_{2019}, Y_{2020})_{20}$	

Table 2: Simplified representation of our empirical setting relative to Chernozhukov et al. (2023).

Therefore, we do not have to split the controls observed in 2018 in two halves to avoid overfitting and this enables us to reconstruct a counterfactual outcome of no treatment for each 2019 exporter without incurring in overfitting problems. Therefore, in this paper for the SUM we use as an auxiliary sample all the Colombian exporters observed in 2018 (A) and as the main sample (M) all the Colombian exporters observed in 2019. For the SAM, we perform instead a K-Fold splitting in which, iteratively we select 80% of the firms in 2019-2020 as being part of A and the remaining 20% as being part of M. This is shown in the column "Our Setting (SAM)" of Table 2, where different A (and, accordingly different M) groups are selected according to the different colors of the dashed circles. In this way we avoid overfitting problems and, at the same time, we exploit all the available data by being able to compare the predicted probabilities to export in the COVID-19 with those in the non-COVID-19 scenario for all the observed 2019 exporters.

Lemma D.1 of Chernozhukov et al. (2023) provides a theoretical foundation for conducting valid inference on key features of CATE. This lemma relies on several underlying assumptions.

[15] Furthermore, even if we assume that during the first three months of the year there was no COVID-19 effect going on, and therefore we categorize as non-treated (treated) firms operating in those months (in the other remaining months), and we use the non treated firms in the auxiliary sample to learn, it would be problematic to use the learning outcome in case of no treatment during the first three months to predict the outcome in case of no treatment for the treated firms that are those in the last 8 months because of the strong seasonality effects we have. So the outcome during the first three months in case of no treatment would be very different from the outcome of the last months in case of no treatment just because of seasonality effects.

Most notably, Chernozhukov et al. (2023) assume a randomized controlled trial (RCT) setting. In our case, this assumption is not overly restrictive, as the estimated distribution of the propensity score in our sample lies within a narrow range of 0.498 to 0.511, closely approximating random assignment according to the distribution of observables (see Figure Appx.3 in Appendix D). Another possible limitation of our setting is the overlap of firms between $t-1$ and t, given by the panel nature of the dataset, which could introduce overfitting problems in our strategy. The original sample splitting procedure of Chernozhukov et al. (2023) instead is not affected by this problem, using different sets of firms in the A and M samples.

The SPE (Chernozhukov et al., 2018) offers formal inference guarantees. In particular, the bootstrap-based confidence bands are valid under mild regularity conditions. We verify these conditions in our setting following the guidance in Chernozhukov et al. (2018), which centers on the smoothness and non-degeneracy of the estimated treatment effects function, as shown in the Online Appendix. However, the SPE is a method originally designed for doing inference on CATEs obtained with parametric and semiparametric estimators, and not for machine learners. In Chernozhukov et al. (2023), Group Average Treatment Effects (GATES), which have been designed for studying CATE estimates obtained with ML methods, have the same role as SPE, as detailed in section 4.5. The main difference between SPE and GATES is that the former summarises the distribution of CATEs by estimating its percentiles, while the latter divides the support of estimated CATEs in bins by typically using quartiles or quintiles and estimates averages of the CATEs within these bins. Finally, the classification analysis (CA) employed in Chernozhukov et al. (2018) and in this paper is the same as the Classification Analysis (CLAN) presented in Chernozhukov et al. (2023).

Therefore, as a robustness check, in section 4.5 we will follow the procedure outlined by Chernozhukov et al. (2024) to apply the Chernozhukov et al. (2023)'s methodology to a research design characterized by the Conditional Independence Assumption and we compare the results obtained with those conveyed by SPE.

3 Data and Dependent Variable

This study focuses on the social and economic disruption caused by the COVID-19 pandemic and its effect on Colombian exporters. This global health crisis triggered by the COVID-19 pandemic served as a notable example of a large-scale economic shock that profoundly impacted global trade, with the dynamics of exporters in Colombia being significantly affected. Applying our ML strategy to data collected from Colombian exporting companies during this period can provide key insights into how companies adapt and survive in the face of such widespread disruption. This provides an understanding of market resilience and firm survival dynamics in the context of global trade shocks. Therefore, our dependent variable $Y_{i,t}$ is a dummy variable that takes the value 1 if a firm i is an exporter at time t –given

that it was an exporter in $t-1$– and the value 0 if the firm is not exporting at time t.[16] By grounding our research on a specific case study, we maintain its relevance to the specific scenario while preserving its potential for broader applications.

We use monthly export transaction data reported at the Colombian Customs Office (Dirección de Impuestos y Aduanas Nacionales, DIAN) for 2018, 2019, and 2020. For each transaction, we consider the exporter ID as the firm identifier; the date; a 10-digit Harmonized System code (HS) characterizing the product; the product origin within Colombia (department level); the means of transportation of the shipment; the country of destination; and, the free on board value of the export transaction in US dollars. This data set also contains information about the value and origin country from which a given exporter imports. We remove all transactions related to re-exports of products elaborated in other countries. As a result, we ended up with 386,132 customs reports in 2018 (7 741 firms), 402,140 in 2019 (7 831 firms), and 365,626 in 2020 (7 518 firms).

3.1 Control Variables

The selection of control variables is based on the determinants of firm entry and exit in foreign trade (see, e.g., Albornoz et al., 2012; Arkolakis et al., 2021). We classify products at the six-digit level of the HS code. We consider different features of exporters according to their monthly exports: the total export (and import) value, the number of products (NP), the number of export destinations (ND), the number of import origin countries (NO), the Herfindahl-Hirschman indexes at the product level (HH_p) and the destination level (HH_d), and a set of dummies for the destinations and origin countries and continents. We create a set of dummies according to the Colombian department from which the product comes, a set of dummies for the means of transportation used, and a set of dummies classifying the product HS-chapter and HS-section. Moreover, we build two sets of dummy variables indicating whether a firm has experience exporting in specific destinations and product sectors. We also account for the accumulated exporting (importing) experience by summing up the total value exported (imported) during the last twelve months. Furthermore, we create four *size* dummies classifying firms according to the quartiles of the firm-level distribution of the total monthly log-value of exports.

To measure the COVID-19 demand and supply shock, we use the information on government contention measures coming from Hale et al. (2020), which consists of four indexes (ranging from 0 to 100) representing the strength of the measures taken by countries to contain the COVID-19 outbreak. The authors provide an economic index summarizing economic policies (E), a health index summarizing health policies (H), a government index describing the strictness of 'lockdown style' policies (G), and an overall government response index called stringency index (S). The value of these indexes ranges from 0 to 100.[17] We

[16] More precisely, $Y_{i,t} = 0$ if a firm is no longer active or is active but not exporting at time t.

[17] These indexes are released daily. We average this information at the monthly level.

build two variables at the firm level for each of the four indexes, one at the export and one at the import side, by taking a weighted average of the country-level scores according to the proportion of the total monthly value of exports (imports) that a firm ships (source) in each country in 2019. We call these firm-level indexes for a firm i "Containment Index$_{i,j,z}$", with $j = \{E, H, G, S\}$ and $z = \{\text{Imp}, \text{Exp}\}$.[18]

Our final data set is composed of 1,975 covariates. They are presented in detail in Table Appx.1 of Appendix A.

4 Results

4.1 Selection of the machine learning algorithm

We evaluate and contrast the outcomes of several ML techniques against a benchmark logistic regression, aiming to identify the model with superior prediction performance, which is crucial for the consistency of our T-Learner estimator. The out-of-sample predictive efficacy of our empirical models is crucial, given our goal to reconstruct an unobserved counterfactual. The complexity of this task arises from its high dimensionality and complex interdependencies between firms and products from various sectors and export destinations. While an approach focusing on in-sample prediction accuracy might overfit, ML techniques optimally balance the bias-variance trade-off for out-of-sample predictions.[19]

We examine four distinct models: Logit, Logit-LASSO, Logit-Ridge, and Random Forest (RF). [20] For Logit, Logit-Ridge, and Logit-LASSO models we include interactions between the *size* of the company and some of the main product characteristics, *industry*, *sector*, *means of transportation* as well as with *destination country* dummies. Notice that Random Forest uses the variables sequentially and, therefore, with a large enough number of trees, it is not necessary to explicitly introduce interactions as explanatory variables, i.e., the model automatically takes into account the interactions that are useful to accurately predict the outcome.[21] The prediction analysis is repeated for all months between January-December 2020.[22]

[18] The value of the Containment Stringency Index Import for firms that are not importing corresponds to the value of the Containment Stringency Index for Colombia (as firms are sourcing all their inputs domestically).

[19] Hyperparameter tuning through cross-validation or other theory-driven methods is often critical in order to avoid overfitting.

[20] Note that it is important to optimize (tune) the hyperparameters of the models for an accurate prediction. These are the hyperparameters we tuned in our models: Ridge; $[10^{-4}, 10^{2}]$, best 10^{0}; Logit-LASSO; $[10^{-4}, 10^{2}]$ best 10^{1}; RF, following Probst et al. (2019); n estimators: $[100, 200, 500]$, best 500; max features: ['sqrt', 'log2', None], best sqrt; max depth: $[5, 7, 10]$, best 7; max leaf nodes: $[3, 6]$, best 6; min samples split: $[2, 8]$, best 2.

[21] For more information about all the features included to build the SUM and SAM see Table Appx.1 in Online Appendix A.

[22] We performed a series of robustness tests for alternative ML methods (XGBoost and SVM) and panel cross-validation (Fabra et al., 2022). The results confirm those presented in the main text and are available upon request.

Table 3 shows the goodness of fit of the model's predictions using two widely used classification metrics: Root Mean Square Error (RMSE) and the Area Under the Receiver Operating Curve (AUC). The best value for the RMSE is 0, which indicates optimal accuracy with no fixed upper limit. The AUC reaches a value of 0.5 for random predictions and 1 when the outcomes are classified without error. Our preferred metric is the RMSE. The reason is twofold. First, our analysis focuses on estimating *predicted probabilities* of exporting, rather than producing binary classifications. The key object of interest is the probability of continuing to export in the future, conditional on covariates and treatment status, which is central to counterfactual analysis. In this setting, thresholding probabilities to assign binary labels (e.g., classifying 0.51 as 1 and 0.49 as 0) can lead to substantial misrepresentation of the underlying uncertainty, particularly in marginal cases. Second, model performance in our context is evaluated based on how well the predicted probabilities approximate the true (unobserved) probabilities. The AUC-ROC is provided as an alternative threshold-independent measure of the quality of the fit. The results are consistent when using different measures of goodness of fit.

The table's upper part displays the accuracy of predictions for the probability of exporting in 2019 based on 2018 exporter data, serving as an out-of-sample performance benchmark in a pre-COVID-19 context using cross-validation. Here, the Logit-LASSO and RF models arise as top performers. The table's middle section also shows the accuracy of models estimated using the exporters' characteristics in 2018 to explain their observed outcomes in 2019; however, these models are now tested using the set of exporters of 2019 and their observed outcomes in 2020. If the functions f_t^0, which represent the relationship between the explanatory variables and the outcome without the pandemic, are sufficiently similar for the year before the pandemic and for the year 2020 (f_{2019}^0 and f_{2020}^0; see assumption (ii)), we expect the accuracy of $\hat{f}_{2019}^0$ to be similar in the first three months of 2020 (when there is likely no relevant COVID-19 effect in Colombia) as in the same months of 2019. Indeed, as expected, the accuracy of Logit-LASSO and RF in January, February and March remains unchanged compared to the accuracy found in the upper part of the table. However, in the middle part of the tables, a decrease in accuracy can be observed after April, highlighting the challenges of a model not trained on COVID-19 data when forecasting in an environment affected by COVID-19.

The models in the lower part of Table 3 are trained and tested with the universe of exporters in 2019 and their observed outcomes in 2020. We use these models to create the SAM forecasts. The accuracy of the predictions is very similar to that obtained with the SUM for 2019 and for the first three months of 2020. In our analysis, it is crucial to achieve accurate predictions because the unbiasedness of our treatment effect estimators depends on the quality of the (counterfactual) prediction accuracy. Both the SUM and the SAM show an acceptable level of accuracy when predictions are made with Logit-LASSO and Random Forest.

Table 3: Goodness of Fit for SUM and SAM in 2018/19 and 2019/20

	RMSE				AUC			
	Logit-LASSO	Logit-Ridge	Random Forest	Logit	Logit-LASSO	Logit-Ridge	Random Forest	Logit
Goodness of Fit for SUM in 2018/19								
Jan	0.42	0.48	0.40	0.59	0.82	0.55	0.83	0.65
Feb	0.44	0.47	0.40	0.59	0.80	0.60	0.82	0.61
Mar	0.41	0.48	0.40	0.59	0.82	0.63	0.82	0.63
Apr	0.40	0.48	0.40	0.58	0.83	0.60	0.83	0.61
May	0.41	0.48	0.40	0.59	0.82	0.57	0.82	0.62
Jun	0.41	0.48	0.40	0.58	0.82	0.57	0.82	0.62
Jul	0.44	0.48	0.40	0.58	0.82	0.64	0.83	0.64
Aug	0.44	0.47	0.40	0.58	0.81	0.60	0.82	0.64
Sep	0.45	0.47	0.40	0.58	0.82	0.73	0.83	0.62
Oct	0.41	0.48	0.40	0.58	0.82	0.62	0.83	0.63
Nov	0.41	0.48	0.41	0.59	0.82	0.67	0.82	0.60
Dec	0.44	0.47	0.40	0.59	0.81	0.61	0.82	0.60
Goodness of Fit for SUM in 2019/20								
Jan	0.42	0.48	0.41	0.59	0.83	0.80	0.83	0.62
Feb	0.45	0.48	0.41	0.59	0.80	0.76	0.81	0.63
Mar	0.41	0.48	0.40	0.57	0.82	0.81	0.83	0.69
Apr	0.48	0.53	0.48	0.60	0.78	0.77	0.79	0.64
May	0.46	0.52	0.46	0.59	0.81	0.80	0.82	0.65
Jun	0.44	0.50	0.43	0.56	0.81	0.78	0.81	0.69
Jul	0.46	0.49	0.42	0.57	0.81	0.79	0.81	0.67
Aug	0.46	0.49	0.42	0.59	0.80	0.79	0.81	0.64
Sep	0.46	0.48	0.41	0.58	0.81	0.78	0.81	0.60
Oct	0.42	0.49	0.42	0.58	0.80	0.78	0.80	0.60
Nov	0.41	0.48	0.41	0.58	0.82	0.79	0.82	0.60
Dec	0.45	0.48	0.41	0.57	0.81	0.79	0.81	0.61
Goodness of Fit for SAM in 2019/20								
Jan	0.44	0.48	0.41	0.59	0.82	0.58	0.82	0.64
Feb	0.46	0.44	0.42	0.60	0.79	0.77	0.80	0.61
Mar	0.41	0.43	0.40	0.59	0.82	0.80	0.83	0.64
Apr	0.42	0.48	0.42	0.60	0.82	0.78	0.80	0.63
May	0.41	0.49	0.41	0.60	0.83	0.78	0.83	0.64
Jun	0.42	0.47	0.43	0.60	0.82	0.78	0.80	0.64
Jul	0.46	0.49	0.42	0.60	0.80	0.71	0.80	0.63
Aug	0.42	0.48	0.42	0.60	0.81	0.64	0.81	0.61
Sep	0.42	0.48	0.42	0.59	0.80	0.66	0.80	0.62
Oct	0.46	0.46	0.42	0.59	0.81	0.77	0.80	0.64
Nov	0.45	0.44	0.41	0.57	0.81	0.78	0.81	0.63
Dec	0.42	0.43	0.42	0.58	0.81	0.78	0.80	0.62

Notes: Results are obtained based on a 5-fold cross-validation strategy. RMSE and AUC are averaged across folds.

4.2 Evaluation of the COVID-19 effect

Both Logit-LASSO and Random Forest reach high accuracy levels in the export status prediction. As explained in section 2, the predicted probabilities are used to estimate the average monthly effect of the COVID-19 shock as the (monthly) average of $\hat{\alpha}_i$ (the difference between the firm-level probabilities of success predicted by the SUM and the SAM.), $\bar{\hat{\alpha}}$. They are presented in Figure 1.

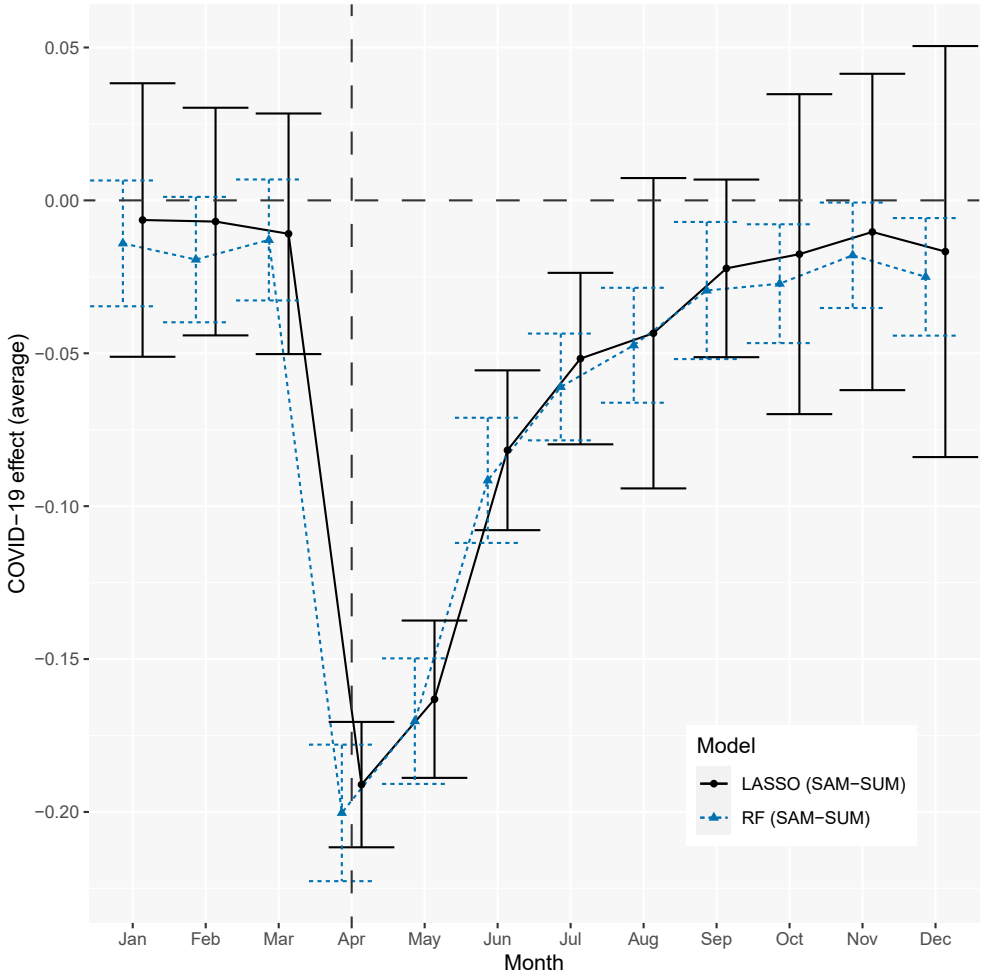


Figure 1: Average Individual Treatment Effect, by months, comparing Logit-LASSO and RF. Standard errors were obtained with 100 bootstrap replications. Confidence intervals for a 5% significance level.

Given the presumption that firms suffered a negligible COVID-19 shock impact during the initial three months of 2020, the treatment effect estimates for this period can be viewed as a placebo test, reminiscent of the in-time placebo test routinely employed in Synthetic Control Methods (Abadie et al., 2015). Detecting a significant COVID-19 effect in the months preceding the actual economic shock would suggest that our model mechanically estimates a COVID-19 effect even in the absence of the stated shock. We conduct these placebo studies also conditioning on exogenous firms' characteristics observed in 2019 by estimating COVID-19 effects for selected subsamples of firms according to such characteristics. We

interpret these additional placebo studies as a robustness check on our results on treatment effect heterogeneity.

As shown in Figure 1, the probabilities obtained from the SUM and the SAM are almost identical on average for January, February, and March. This result is reassuring since only from March 25, 2020, the Colombian government implemented a complete and mandatory lockdown.[23] More in general, we can conclude that our identification strategy is not mechanically recovering COVID-19 effects for a period with low incidence in Colombia and in the rest of the world. We find that the peak of the COVID-19 effect is in April 2020, when we estimate an average difference between the predicted probabilities of exporting of nearly 20 percentage points. In the following months, the estimated average effect declines with time.

The results indicate that both Logit-LASSO and RF models yield comparable performances.[24] Given their good performance and considering that Logit models are frequently used in similar contexts, we opt for Logit-LASSO. It aligns with the conventional approaches and offers greater interpretability as an extension of the traditional model.[25]

The results obtained using the AIPW-Double ML estimator (with 5-fold cross-fitting and nuisance parameters estimated with Random Forest), which are shown in Figure Appx.2 in the Appendix, are equivalent. It is also interesting to notice that estimated ATT and $ATE_{\{t_s,t_s-1\}}$ are practically the same. This happens because the distribution of the explanatory variables is exactly equal between the treated and the control group, as it is shown in Figure Appx.3 that reports the results of the estimated propensity score for the two groups.

Figure 2 reports the estimated CATE_z by industry, that is the Conditional Average Treatment Effect for those units belonging to different industries. Concentrating on the results for the first quarter, we are able to reject the existence of an effect even within sectors.[26] During the second quarter of 2020, Colombian exporters belonging to almost every industry are found to significantly reduce their probability of surviving in the international markets. This decline is particularly pronounced in industries such as Textiles and Footwear, and Jewelry. In line with the aggregate picture, the estimated effects decrease in magnitude in the third quarter and are no longer statistically significant in the fourth quarter.

[23] See Online Appendix A for a detailed description of the measures taken in Colombia in the midst of the COVID-19 crisis.

[24] This is somewhat expected since in Künzel et al. (2019) meta-learners are said to provide better outcomes with generalizable ML-algorithms that perform well for a large variety of data sets.

[25] Non-reported results using RF are equivalent and available upon request.

[26] We have conducted other similar placebo studies conditioning on other variables (e.g., the main destination of exports, the main origin of imports, via (air, land, sea), industry, exported value, imported value) and in all the considered subsamples we do not estimate any significant effect of COVID-19.

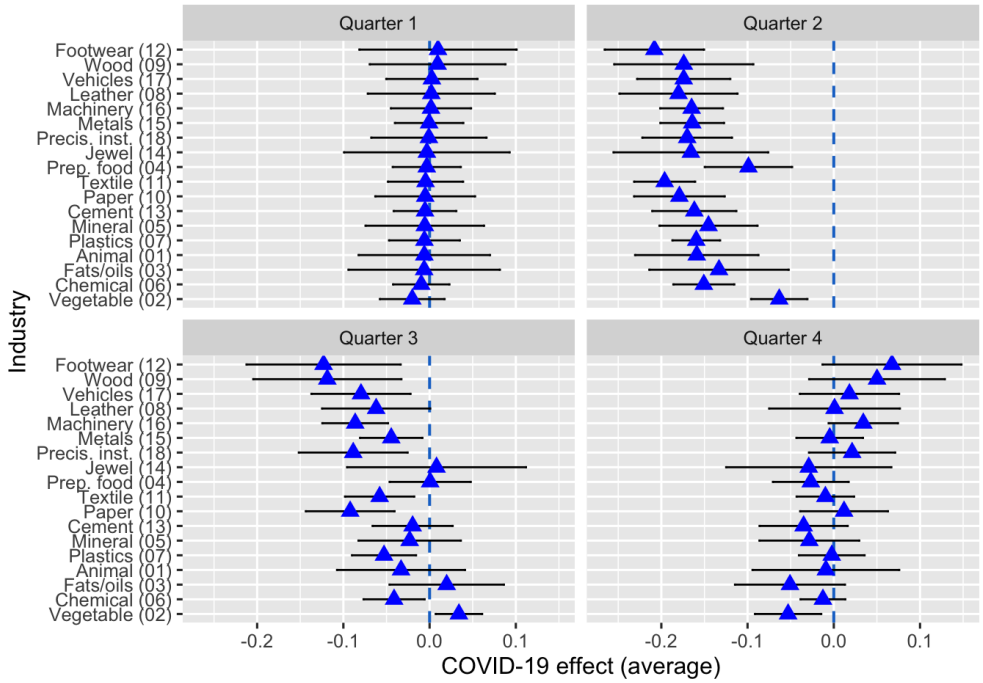


Figure 2: The quarterly mean difference in the predicted probability of success (SAM vs. SUM) by industry, using the Logit-LASSO predictions. Standard errors were obtained with 100 bootstrap replications. Confidence intervals for a 5% significance level.

4.3 Heterogeneity of the COVID-19 effect on Colombian exporters

In this section, we investigate the determinant of possible treatment effect heterogeneity. Figures 3 and 4 show the estimated Sorted Partial Effects (SPE) and Average Partial Effects (APE), which are obtained as explained in section 2 by month and aggregating all the months, respectively. The two figures also report the 95% confidence intervals with blue bands for SPE and black dashed lines for APE.

In the pre-pandemic months, the SPEs closely aligned with the APEs (estimated to be zero). This demonstrates that individual placebo treatment effects are not statistically significant throughout the distribution, not just on average, reinforcing the robustness of our methodology across the entire distribution of treatment effects. By contrast, significant treatment-effect heterogeneity is observed in the remaining months, except for June and December.

To identify the determinants of treatment effect heterogeneity, we examine the difference in means ($CADiff$) of firm characteristics between the most and least affected groups in Table 4. These groups are defined by whether their estimated α_i is lower than $\alpha^*(25th)$ or greater than $\alpha^*(75th)$, respectively. Therefore, we compute the raw difference in the means of the covariates between the most and the least affected firms by regressing the variables of interest on a constant and a dummy $q = \mathbf{1}_{\{\alpha_i \leq \alpha^*(25th)\}}$ for the observations for which estimated $\alpha_i \leq \alpha^*(25th)$ or $\alpha_i \geq \alpha^*(75th)$. Then, we also provide the difference in adjusted means once we have controlled for the firm sector and month of the year. Controlling for

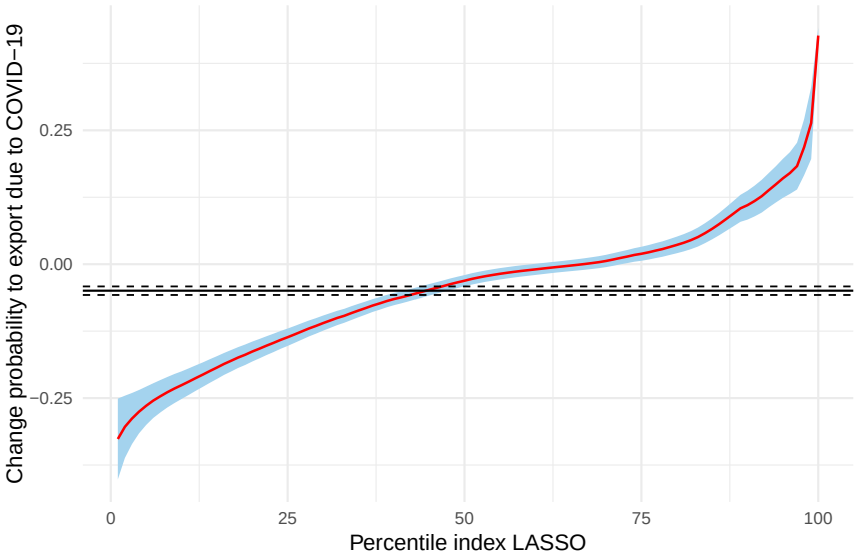


Figure 3: Annual Sorted Partial Effects (SPE) and Average Partial Effects (APE) of COVID-19 on Colombian firms' export status. The treatment effect is calculated as a difference between SAM and SUM predictions. Standard errors were obtained with 100 bootstrap replications. Confidence intervals for a 5% significance level.

sector and month allows us to perform a *ceteris paribus* analysis, i.e., to dig into the effects of COVID-19 within specific sectors and specific months.

Table 4 is divided into 3 columns according to the control variables included in the regressions: in the first column, we show the unconditional average difference in the firms' characteristics between the most and least affected firms; in the second column, we control for the firm industry; and, in the third column, we control for firm industry and month of observation. The firm characteristics that we consider to explore the sources of COVID-19 treatment effect heterogeneity among Colombian exporters are observed in 2019 (the year before receiving the treatment). First, we check whether the estimated individual treatment effect (TE) differs between the firms contained in the two groups by using the TE as the dependent variable. We then move to firm-sector specific characteristics. In particular, the first set of firm characteristics that we use as dependent variables are dummies indicating the industry where the exporters operate.[27] We also investigate the *CADiff* for the means of transportation and the months when firms operate. Moreover, to account for the role of diversification patterns, we also consider as dependent variables the number of export destinations (ND), import origins (NO), and products (NP) exported. The weighted Containment Stringency Index that exporters face when exporting (importing) allows us to

[27]We aggregate the 22 industries defined in the main analysis as follows. "Agriculture" contains Animals (01), Vegetables (02), Fats/oils (03), and Prepared Foodstuffs (04). "Chemicals" includes Chemical (06), and Plastics (07). "Manufacturing" contains Machinery (16), Vehicles (17), and Manufactured (20). "Metals" aggregates Mineral (05), Cement (13), Jewelries (14), and Metals (15). "Special" includes Precision Instruments (18), Arms (19), Art (21), and Special (22). "Textile" contains Leather (08), Textile (11), and Footwear (12). Finally, "Wood" aggregates Wood (09), and Paper (10). See Table Appx.2 in the Online Appendix for the complete industry names.

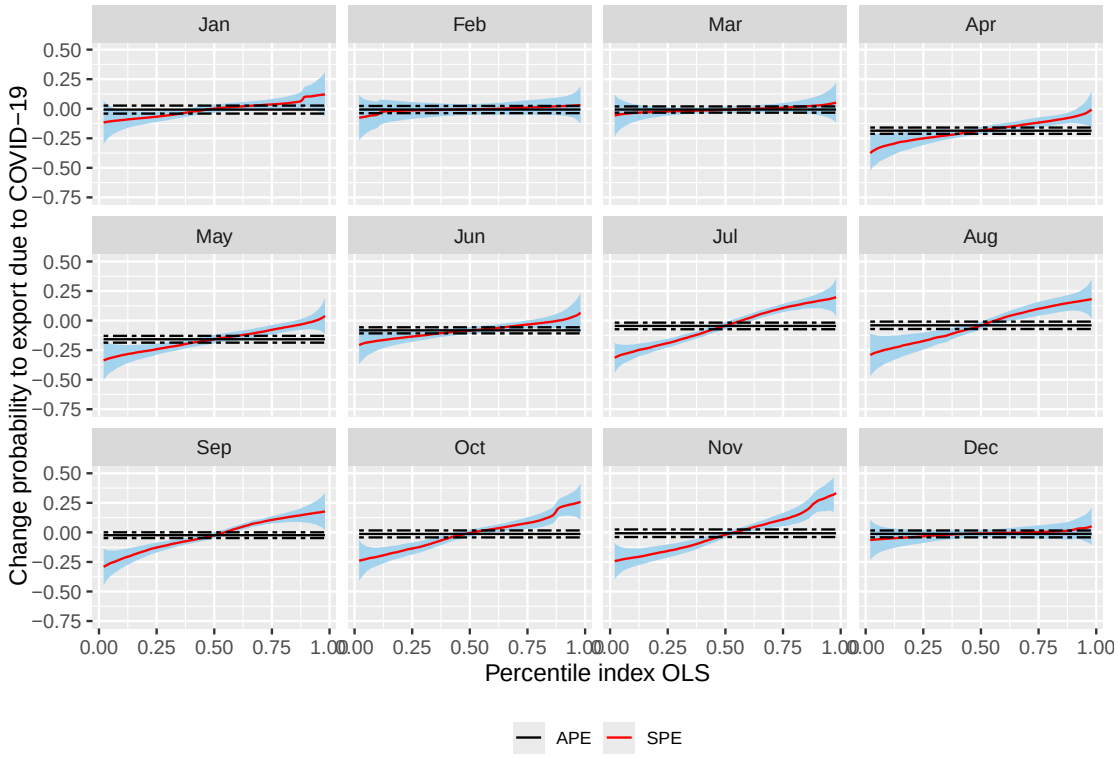


Figure 4: Monthly Sorted Partial Effects (SPE) and Average Partial Effects (APE) of COVID-19 on Colombian firm export status. The treatment effect is calculated as a difference between SAM and SUM predictions. Standard errors were obtained with 100 bootstrap replications. Confidence intervals for a 5% significance level.

study to what extent treatment effect heterogeneity depends on these firm-specific measures of exposure to COVID-19 through their activities on international markets. A traditional continuous-DID strategy would have used these intensity variables as treatment variables, assuming that any COVID effect would emanate through them. Finally, including the total value exported (imported) by firms –expressed in logarithm– among the variables for which the *CADiff* is computed highlights the difference in the quantities sold (purchased) by most and least affected companies. A discussion of the main findings follows.

Considering the estimated individual treatment effects (TE) as a dependent variable, we find a negative and significant difference between most and least affected firms independently of the set of controls employed. These results show that the most affected exporters–i.e., those located in the first SPE quartile distribution–experienced a decrease in the probabilities of exporting 32 percentage points lower than the one experienced by the least affected firms–i.e., those located in the last SPE quartile.

We found significant differences among firms when examining how different aggregate sectors are affected. For instance, we detect that the share of textile firms among the most affected 2019 exporters is approximately 5 percentage points higher with respect to the one estimated for the group of the least affected firms. Likewise, there is a difference of 2 percentage points in the presence of exporters belonging to the Chemicals industry between

Table 4: Estimated differences in means of the estimated treatment effect and other covariates between the group of more affected and the group of less affected firms ($CADiff$) applying the classification analysis to the $SAM - SUM$ estimates

Outcome variable	(1)	(2)	(3)
TE	-0.320^{***}	-0.318^{***}	-0.325^{**}
Agriculture	-0.118		
Chemicals	0.0229^{*}		
Manufacturing	0.019		
Metals	0.009		
Special	0.003		
Textile	0.049^{***}		
Wood	0.0143		
Air	0.0351^{*}	0.0391^{*}	0.0277
Land	0.0376	0.0246	0.00564
Sea	-0.0718^{*}	-0.0637^{***}	-0.0334
Jan	-0.114^{***}	-0.119^{***}	
Feb	-0.043^{**}	-0.042	
Mar	-0.052^{***}	-0.052^{***}	
Apr	0.223^{***}	0.221^{***}	
May	0.184^{***}	0.183^{***}	
Jun	0.056^{*}	0.056^{***}	
Jul	-0.018	-0.015	
Aug	-0.027	-0.024	
Sep	-0.065^{***}	-0.063^{**}	
Oct	-0.062^{**}	-0.062^{***}	
Nov	-0.043	-0.042^{*}	
Dec	-0.037	-0.038	
Number of export destinations (ND)	-0.703	-0.500^{*}	-0.280
Number of import origins (NO)	-1.7470	-1.9820^{***}	-2.4440^{**}
Number of exported products (NP)	-0.313	-0.542^{*}	-0.616^{*}
Containment Index Stringency Export	27.28^{**}	25.970^{***}	15.150^{***}
Containment Index Stringency Import	2.760^{***}	2.073^{**}	-0.542
Value Exported (log)	-0.622	-0.521^{***}	-0.730^{***}
Value Imported (log)	-0.558	-0.866^{**}	-0.934^{*}
Deviation from industry mean		✓	✓
Deviation from monthly mean			✓

Notes: column (1) does not include industry or month variables in the regression; column (2) includes industries in the regression; and, column (3) includes both the industry and month variables. *** means significant at 1%, ** at 5%, and * at 10%. Standard errors are obtained by bootstrapping the whole estimation process, and joint p-values are adjusted to consider the simultaneous testing of all variables.

the most and least affected groups.

We also detect the existence of treatment effect heterogeneity associated with the means of transportation used by exporters in 2019. On the one hand, there are approximately 4 percentage points more exporters using air transportation among the most affected than among the least affected firms. On the other hand, there are approximately 7 percentage points fewer Colombian exporters using the sea for shipping among the most impacted firms compared to the least affected ones (Nitsch, 2022). However, these differences become not statistically significant once we control for the month of observation.

Looking at treatment effect heterogeneity by month, the most relevant finding is that, consistently with the previous results, only April, May and, once industry is controlled for,

June show economically meaningful positive estimates that are statistically significant at the 1% level. There are approximately 22 percentage points more firms in April among the most affected than among the least affected firms, with corresponding differences of 18 and 6 percentage points for May and June, respectively. For the other months, the coefficients tend to be negative or statistically insignificant.

To evaluate whether ex ante exporter diversification shaped the impact of the COVID-19 shock, we examine the estimated parameters associated with ND, NO, and NP. Specifically, we investigate whether Colombian exporters' supply-chain diversification and export-destination diversification helped mitigate the effects of the shock. Without controlling for industry affiliation, we do not find compelling evidence that ex ante diversification played a significant mitigating role, as shown by the statistically insignificant estimated parameters associated with ND, NO, and NP in column (1). However, once we control for industry, and thus, inter alia, for industry-level differences in diversification potential, the estimated negative differences become statistically significant. This suggests that, within industries, firms that were less diversified before the shock were more exposed to its adverse effects. The most striking effect, statistically significant at the 1% level, is found for NO. Within industries, the most affected Colombian exporters imported from 1.98 fewer countries in 2019 than the least affected firms. The magnitude of this estimate is economically sizable, given that approximately 60 percent of Colombian exporters are not integrated into global value chains, since they do not import, and that the mean of NO is approximately 4.16 import origins. Finally, the most affected Colombian exporters exported to 0.5 fewer countries and 0.5 fewer products. However, the estimated coefficients for ND and NP are only marginally statistically significant, at the 10% level, and are less economically relevant, given that the means of ND and NP are 2.6 and 3.4, respectively.

The $CADiff$ estimates obtained using the Export and Import Containment Stringency Indices as dependent variables provide useful insights into the difficulties faced by Colombian firms when exporting to, or importing from, countries that adopted severe stringency measures. As expected, the most affected Colombian exporters were, on average, exposed to higher Export and Import Containment Stringency Indices than the least affected firms.[28]

Finally, as expected, Colombian exporters trading larger volumes, in value terms, appear to be more resilient to the COVID-19 shock. As with diversification, the comparison between export and import sides suggests that sourcing inputs from abroad is associated with greater attenuation of the shock.

Our results not only show the uneven impact of the COVID-19 crisis across different covariates, but also highlight the potential of our methodology as a diagnostic tool for targeted policy interventions. By identifying the most affected firms and sectors, our framework can support the allocation of scarce public resources and the design of sector and firm-specific

[28]The effect on the import side becomes statistically insignificant once we control for the month of observation. Recall that the index ranges from 0 to 100.

support programs aimed at improving the resilience of the most vulnerable exporters.

4.4 Estimations based on $Y - SUM$

In this paragraph, following Fabra et al. (2022) and Cerqua and Letta (2020), we use the estimators based on Eq. (13). These estimators capture the differences between the observed outcome, Y (binary variable accounting for the success of a Colombian exporter in 2020), and its counterfactual predictions (SUM). As shown in Figure 5, when the interest lies in estimating the average treatment effects (by months in this case), the results based on $Y - SUM$ do not differ from those obtained by using $SAM - SUM$. We obtain similar results for the two methodologies also in terms of conditional treatment effects based on subgroups defined on firm characteristics (e.g., by industry or main export destination country).

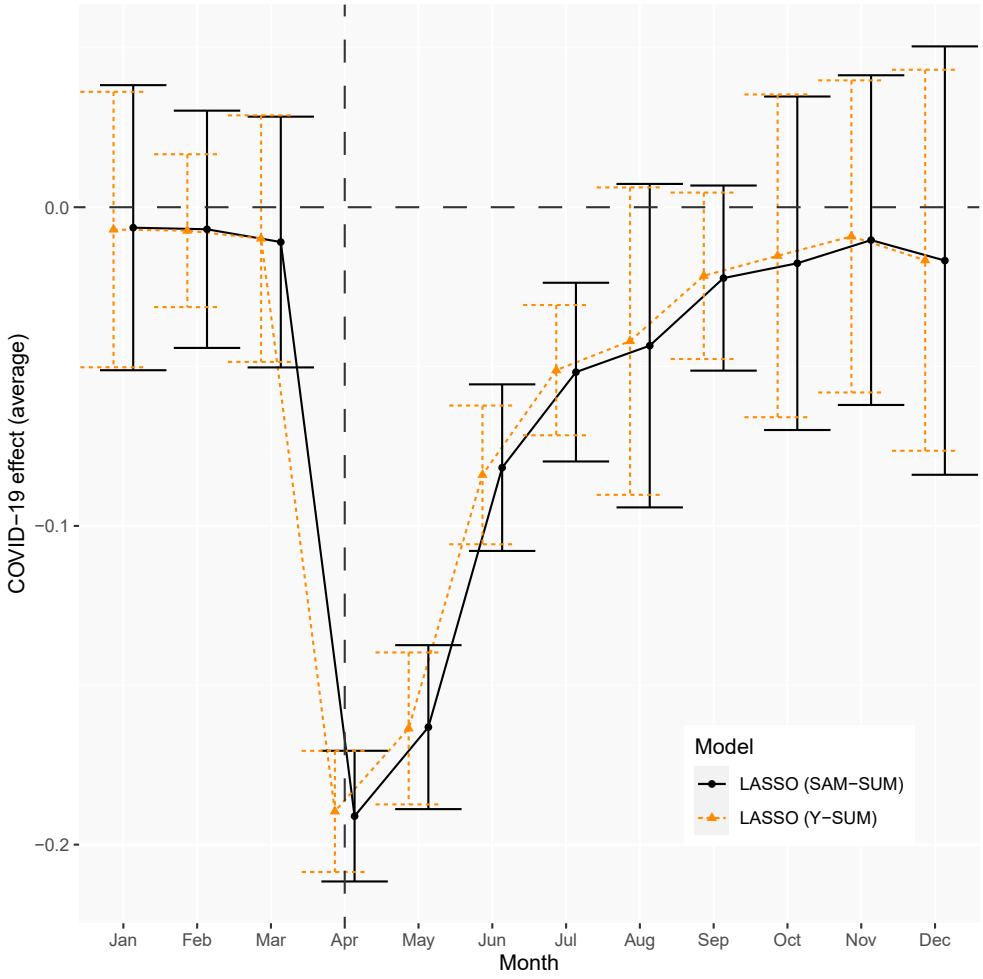


Figure 5: Mean difference in the predicted probability of success by month, comparing $SAM - SUM$ and $Y - SUM$, using Logit-LASSO predictions. Standard errors were obtained with 100 bootstrap replications. Confidence intervals for a 5% significance level.

The fact that the two estimators consistently find zero estimated effects for all 2019 exporters (and for subgroups based on the values of individual observables) during the first quarter suggests that the estimation error of both SUM and SAM, $\mathcal{E}^0$ and $\mathcal{E}^1$ respectively,

goes to zero when we average the individual treatment effects across the whole distribution of 2019 exporters (or in subgroups defined by one of the possible dimension of treatment effect heterogeneity defined by observables; e.g., by industry or main export destination country).

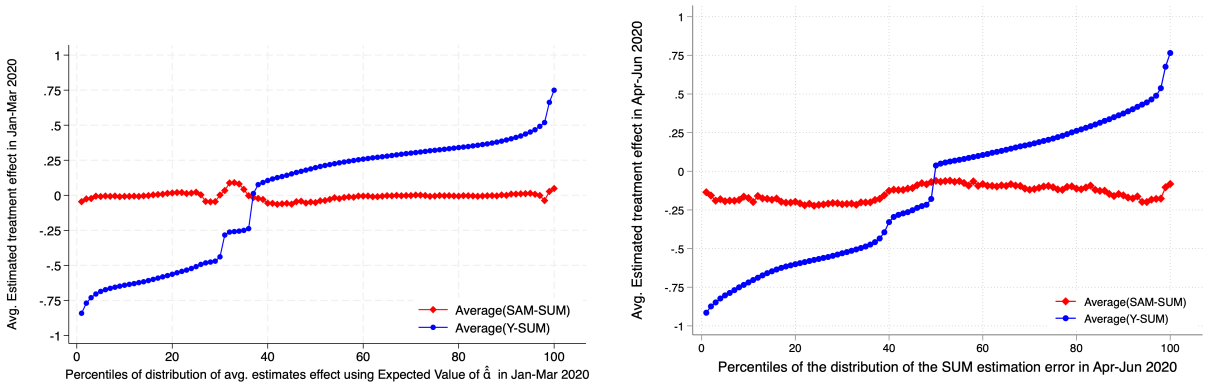


Figure 6: Estimated average treatment effects ($SAM - SUM$, red line, and $Y - SUM$, blue line) for the first and second quarters for observations contained in intervals defined by the estimated percentiles of $Y - SUM$.

However, since our goal is to identify the main dimensions of treatment effect heterogeneity by classifying units with the highest and lowest estimated treatment effects, we need also to evaluate how well these alternative estimation strategies perform in identifying treatment effects at the extremes of the distribution of treatment effects. Figure 6 shows the average of the estimated treatment effects obtained with the two estimators for the observations whose estimated treatment effects (by using $Y - SUM$) are contained in intervals defined by two consecutive values of the estimated percentiles of $Y - SUM$. On the one hand, the estimator based on $Y - SUM$ is also identifying significant treatment effect heterogeneity in the first quarter, suggesting that the distribution's estimation error, $\mathcal{E}^0$, is not zero on average in the tails. Moreover, the shape of the $Y - SUM$ curve is similar across the two quarters, suggesting that this estimation method will be prone to misclassifying units when using the Sorted Effects strategy suggested above. On the other hand, in the first quarter, the shape of the $SAM - SUM$ curve is flat, showing a constant average estimated effect that is zero along the whole distribution of the $Y - SUM$ estimated effects, suggesting that by using the SAM we are able to wash out the estimation error of the SUM because $\mathcal{E}^1 = \mathcal{E}^0$.

This behavior of the estimators based on $SAM - SUM$ is consistent with the results shown in Figure 4 for the Sorted Effects analysis. Figure 7 shows that the intuition on the inadequacy of the $Y - SUM$-based estimators to identify treatment effects on the tail of the distribution is also confirmed by the Sorted Effects analysis based on this estimation strategy. When using the $Y - SUM$ individual level estimates to feed the SPE methodology, we find economically and statistically significant effects of the COVID-19 shock all along the percentile distribution in the first quarter. While it is true that, on average, $\mathcal{E}^0$ tends to be zero across all observations, these findings suggest that this is not true when we focus on specific segments of the treatment effect distribution, particularly in the tails.

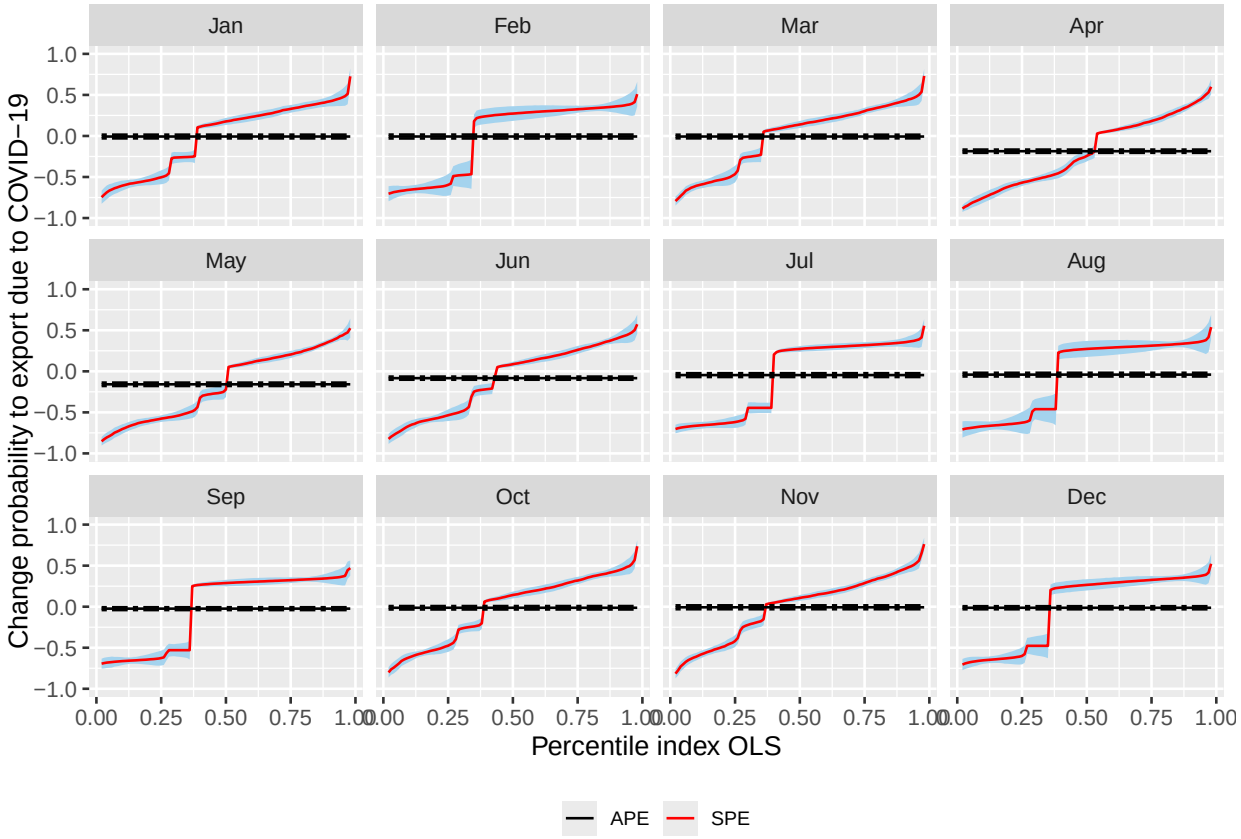


Figure 7: Monthly Sorted Partial Effects (SPE) and Average Partial Effects (APE) of COVID-19 on Colombian firms' export status. TE is calculated as a difference between the observed outcome (Y) and SUM predictions. Standard errors were obtained with 100 bootstrap replications. Confidence intervals for a 5% significance level.

Table 5 presents the classification analysis results on the sources of treatment effect heterogeneity when $CADiff$ is estimated using the $Y - SUM$ approach. Several results contrast with those reported in Table 4 and with what one would expect. Firms that are more diversified on the export side appear to lose relatively more; the coefficients associated with the containment indexes are irrelevant when industry and month controls are omitted, and become negative once these controls are included; and firms exporting larger volumes tend to be relatively more affected. These patterns are consistent with the inability of the $Y - SUM$ approach to estimate treatment effects reliably in the tails of the distribution of α and, consequently, to correctly identify the groups of the most affected and least affected firms. In other words, these groups are likely to be contaminated by firms that are incorrectly classified because of the estimation error $\mathcal{E}^0$.

Table 5: Estimated differences in means of the estimated treatment effect and other covariates between the group of more affected and the group of less affected firms ($CADiff$) applying the classification analysis to the $Y - SUM$ estimates

Outcome variable	$\beta_{1,f}^{(1)}$	$\beta_{1,f}^{(2)}$	$\beta_{1,f}^{(3)}$
TE	-1.058^{***}	-1.060^{***}	-1.058^{***}
Agriculture	-0.078^{***}		
Chemicals	-0.023^{***}		
Manufacturing	0.020^{***}		
Metals	0.034^{***}		
Special	0.004^{***}		
Textile	0.032^{***}		
Wood	0.011^{***}		
Air	-0.021^{***}	-0.020^{**}	-0.017^{**}
Land	0.014^{***}	0.009	0.008
Sea	0.008	0.011	0.010
Jan	-0.021^{***}	-0.021^{***}	
Feb	-0.013	-0.013	
Mar	-0.016^{***}	-0.016^{*}	
Apr	0.050^{***}	0.050^{***}	
May	0.062^{***}	0.061^{***}	
Jun	0.018^{***}	0.018^{***}	
Jul	0.001	0.001	
Aug	0.009	0.009	
Sep	-0.020^{***}	-0.018^{***}	
Oct	-0.026^{**}	-0.026^{**}	
Nov	-0.024^{***}	-0.022^{***}	
Dec	-0.018^{**}	-0.017^{**}	
Number of export destinations (ND)	0.274^{***}	0.298^{***}	0.279^{***}
Number of import origins (NO)	0.0350	-0.0595	-0.1077
Number of exported products (NP)	0.268^{***}	0.162^{***}	0.110^{**}
Containment Index Stringency Export	0.436	0.107	-3.550^{***}
Containment Index Stringency Import	-0.338	-0.916^{*}	-1.536^{***}
Value Exported (log)	0.365^{***}	0.351^{***}	0.328^{***}
Value Imported (log)	-0.426^{***}	-0.623^{***}	-0.674^{***}
Deviation from sectoral mean		✓	✓
Deviation from monthly mean			✓

Notes: column 1 does not include sector or month variables in the regression; column 2 includes sector in the regression, and, column 3 includes both the sector and month variables. *** means significant at 1%, ** at 5%, * at 10%. Standard errors are obtained by bootstrapping the whole estimation process and joint p-values are adjusted to take into account the simultaneous testing of all the variables.

4.5 Validation of the CATE models

As it is well established in the causal inference literature, the Conditional Average Treatment Effect (CATE) coincides with both the Conditional Average Treatment Effect on the Treated (CATT) and the Conditional Average Treatment Effect on the Untreated (CATU) when the conditioning set includes all explanatory variables (those satisfying the conditional independence assumption). This is shown in Figure 8, where we represent the estimated heterogeneous treatment effects for January obtained by using the T-Learner for $CATE_{\{t_s,t_s-1\}}$, $CATU$ and $CATT$. The latter coincides with our estimator $\hat{\alpha}_i$. Finally, we also report the

CATT obtained with $\hat{\hat{\alpha}}_i$.[29]

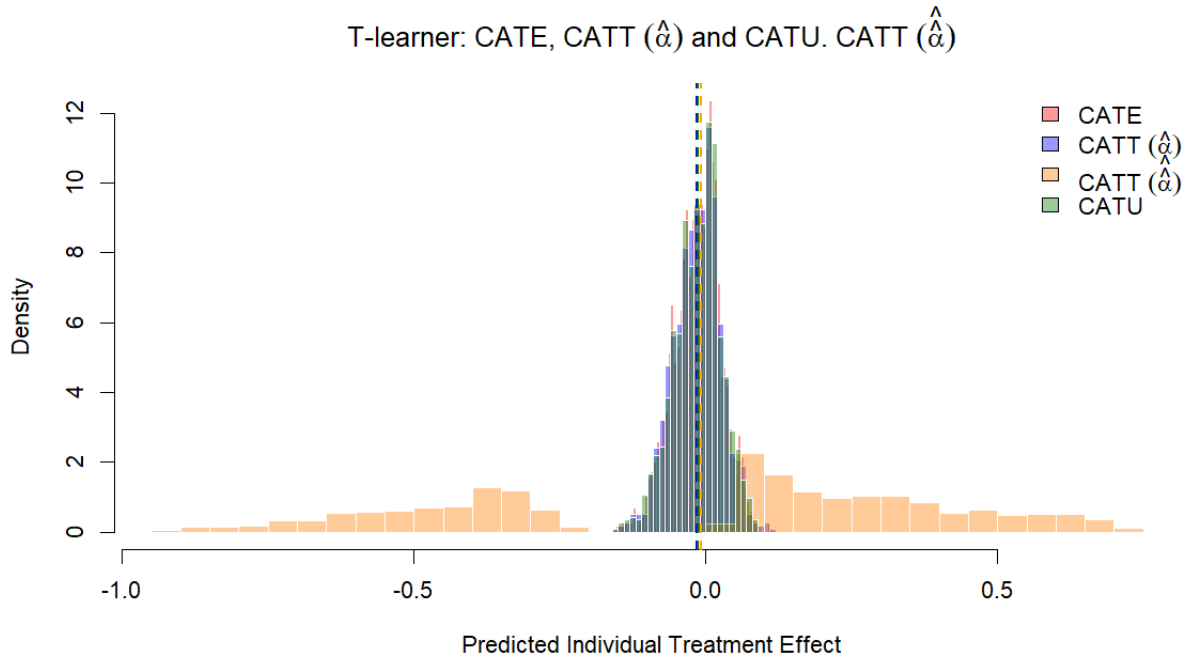


Figure 8: Heterogeneous treatment effects estimated in the A sample and predicted in the M sample for the month of January.

For the four estimators, we use Random Forest to estimate the potential outcomes. In our context, the distribution of observable characteristics is identical for firms that were observed exporting in 2018 (control group) and those exporting in 2019 (treated group). In Figure 8, this implies that the density of the $CATT$, $CATU$ and $CATE_{\{t_s,t_s-1\}}$ estimated with the T-Learner are the same. In this paragraph, we will compare results obtained with the T-Learner and other widely used estimators focusing on the $CATE_{\{t_s,t_s-1\}}$.

The primary aim of this section, which can be interpreted as a robustness check of the previous results, is to follow the procedure described by Chernozhukov et al. (2024) to apply the methodology of Chernozhukov et al. (2023) to a research design characterized by the Conditional Independence Assumption. We will compare the results obtained with SPE (and CA) with those obtained here by using GATES and CLAN. Finally, we will also estimate the Best Linear Predictor (BLP) of the CATE, a methodology to statistically test if a CATE estimator captures any statistically significant indication of treatment effect heterogeneity.

As an additional robustness check, we will also compare the results obtained with our T-Learner with those obtained by using other commonly adopted methods in the literature of causal inference (Chernozhukov et al., 2024), namely: the S-Learner, the R-Learner, the DR-Learner and the Causal Forests (within the Generalized Random Forests framework). An overview of the adopted learners can be found in Appendix B.

[29]Notice that we have exactly the same result obtained in Figure 7 for January: the estimators based on Y–SUM produce a biased estimated distribution of the conditional treatment effects.

Throughout the section, we will randomly divide the observations in an auxiliary (A), consisting in 60% of observations, and a main (M) sample as recommended by Chernozhukov et al. (2023).[30] The models used to identify CATE are estimated using A and are tested in M. Chernozhukov et al. (2023) show that under mild regularity conditions the estimated parameters are normally distributed, conditional on the random sample split on A. To take into account the uncertainty originating from the splitting procedure, they also propose a multisplitting inference strategy in which the researcher repeats the estimation using different random division in splits and report the median of those estimates, along with the medians of the corresponding confidence intervals and p-values.[31]

Following Chernozhukov et al. (2023), we start by statistically testing the capacity of our estimators to capture treatment effect heterogeneity by using the BLP. As previously mentioned, Chernozhukov et al. (2024) propose a strategy to apply their methodology in a non-experimental setting. This approach relies on the pseudo-outcome—computed on the main sample M—of the AIPW-DML estimator (using cross-fitting with 5-fold and Random Forest), which is reported in formula (15).

If the CATE model $\hat{\Delta}(X_{\{t_s-1,t_s-2\}})$ is well-specified, then the best linear predictor of the true CATE using the variables $(1, \hat{\Delta}(X_{\{t_s-1,t_s-2\}}))$ should yield a statistically significant coefficient on $\hat{\Delta}(X_{\{t_s-1,t_s-2\}})$. Given that $\mathbf{E}[\tilde{Y}^{ATE}|X] = \Delta(X)$, the BLP we estimate takes the following form:

$$(\beta_1, \beta_2) = \arg\min_{b_1,b_2} \mathbf{E}\left[\left(\tilde{Y}^{ATE}_{\{t_s-1,t_s\}} - b_1 - b_2(\hat{\Delta}(X_{\{t_s-1,t_s-2\}}) - \mathbf{E}[\hat{\Delta}(X_{\{t_s-1,t_s-2\}})])\right)^2\right]$$

where $\hat{\Delta}(X_{\{t_s-1,t_s-2\}})$ is the CATE estimated in A and predicted in M.

Under regularity conditions, indeed, the coefficient β_2 converges in population to

$$\beta_2 = \frac{\text{Cov}(\Delta(X_{\{t_s-1,t_s-2\}}), \hat{\Delta}(X_{\{t_s-1,t_s-2\}}))}{\text{Var}(\hat{\Delta}(X_{\{t_s-1,t_s-2\}}))}$$

In an ideal case where $\hat{\Delta}(X_{\{t_s-1,t_s-2\}})$ is perfectly aligned with the true CATE, we expect $\beta_2 = 1$. Hence, a statistically significant and positive β_2 provides evidence that the CATE model captures heterogeneous treatment effects.

For each month, Table 6 presents the median of the estimated β_2 coefficients of the CATE across 100 (A, M) splits with the relative median p-value for all the months and the mentioned CATE estimators (we also provide the R^2 of the regression).[32] The results show

[30] Chernozhukov et al. (2023) require that the auxiliary sample be larger than the main sample.

[31] The results of this section using a single split are also available upon request.

[32] Following Chernozhukov et al. (2023), the goodness-of-fit of a CATE estimator $\hat{\Delta}(X_{\{t_s-1,t_s-2\}})$ in the BLP framework is indexed by the signal strength measure $\Xi := |\beta_2|^2 \cdot \text{Var}(\hat{\Delta}(X_{\{t_s-1,t_s-2\}})) = \text{Corr}^2(\Delta(X_{\{t_s-1,t_s-2\}}), \hat{\Delta}(X_{\{t_s-1,t_s-2\}})) \cdot \text{Var}(\Delta(X_{\{t_s-1,t_s-2\}}))$. Maximizing Ξ is equivalent to maximizing the correlation between the estimated and true CATE functions, which in practice corresponds to maximizing

that, from April to August, most learners deliver statistically significant β_2 coefficients, with magnitudes close to one. This suggests that these models are able to detect meaningful treatment effect heterogeneity during these months. By contrast, in the earlier months, January to March, all learners produce small and statistically insignificant coefficients. The only difference with respect to the main analysis is that, in this exercise, the methods do not detect treatment effect heterogeneity from September to November. Overall, therefore, Table 6 confirms the findings from the previous sections.

To evaluate heterogeneity in treatment effects using our estimated models $\hat{\Delta}((X_{\{t_s-1,t_s-2\}}))$, we also use the *Group Average Treatment Effects (GATES)* following Chernozhukov et al. (2023). We slice the distribution of the estimated CATE into K parts and we are interested in the average effect of firms within each slice. Table 7 displays the latter averages. Formally, for a partition of the support of $\hat{\Delta}(X_{\{t_s-1,t_s-2\}})$ into 4 quartile-based groups $G_k := \{\Delta \in I_k\}$, the GATE for group G_k is defined as

$$\gamma_k := \mathbf{E}[\tilde{Y}^{ATE}_{\{t_s-1,t_s\}} \mid G_k].$$

Situations where the γ_ks are similar indicate that no systematic heterogeneity was detected.

The results presented in Table 7 corroborate the findings from the main analysis. In particular, the placebo tests for the first three months of the year are successful (i.e., the estimated GATES are all equal and not statistically significant) and the GATES estimates are substantially larger in magnitude and statistically different during the COVID-19 months—specifically April, May, and June—with some residual evidence in July. In other words, the probability of exporting declined significantly across all subgroups during these months, with the impact varying by group, thereby confirming the presence of meaningful treatment effect heterogeneity. Figure 9 provides a visual representation of the GATES for the different estimators of CATE with the associated confidence level for the first split (which are very similar to the more robust multisplit results in Table 7).

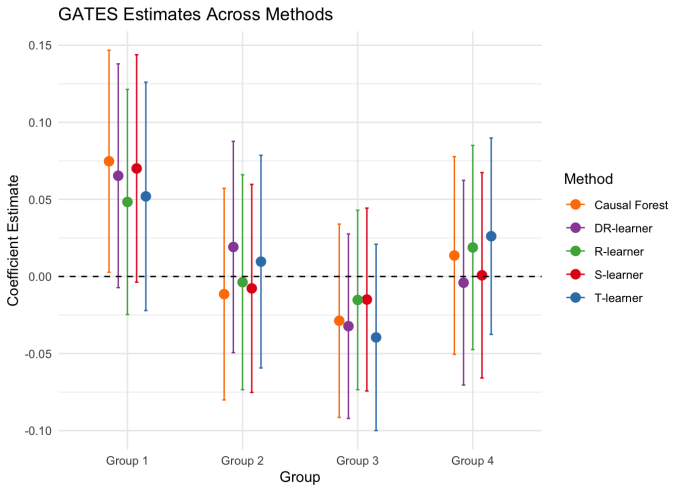
the R^2 of the BLP regression. Thus, the learner with the highest R^2 in a period is the one picking-up an higher degree of treatment effect heterogeneity.

Month	S-learner	T-learner	R-learner	DR-learner	Causal RF
	0.146	-0.254	-0.272	0.042	-0.209
Jan	(0.636)	(0.444)	(0.363)	(0.625)	(0.606)
	$R^2 = 0.00011$	$R^2 = 0.00030$	$R^2 = 0.00040$	$R^2 = 0.00012$	$R^2 = 0.00012$
	-0.236	0.248	0.040	0.183	0.460
Feb	(0.195)	(0.367)	(0.522)	(0.570)	(0.372)
	$R^2 = 0.00078$	$R^2 = 0.00040$	$R^2 = 0.00019$	$R^2 = 0.00015$	$R^2 = 0.00036$
	0.981	0.611	0.048	0.362	0.724
Mar	(0.584)	(0.061)	(0.554)	(0.278)	(0.190)
	$R^2 = 0.00015$	$R^2 = 0.00162$	$R^2 = 0.00017$	$R^2 = 0.00052$	$R^2 = 0.00080$
	2.673	1.119	1.059	0.968	1.541
Apr	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	$R^2 = 0.01350$	$R^2 = 0.01814$	$R^2 = 0.01616$	$R^2 = 0.01361$	$R^2 = 0.01689$
	1.960	1.068	0.992	1.092	1.820
May	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	$R^2 = 0.00581$	$R^2 = 0.01082$	$R^2 = 0.00980$	$R^2 = 0.01134$	$R^2 = 0.01232$
	1.766	0.588	0.634	0.725	1.331
Jun	(0.131)	(0.021)	(0.023)	(0.007)	(0.004)
	$R^2 = 0.00101$	$R^2 = 0.00233$	$R^2 = 0.00226$	$R^2 = 0.00316$	$R^2 = 0.00350$
	0.758	0.738	0.605	0.425	0.836
Jul	(0.607)	(0.017)	(0.029)	(0.157)	(0.176)
	$R^2 = 0.00011$	$R^2 = 0.00251$	$R^2 = 0.00210$	$R^2 = 0.00088$	$R^2 = 0.00077$
	2.022	0.551	0.569	0.537	0.885
Aug	(0.177)	(0.049)	(0.031)	(0.045)	(0.048)
	$R^2 = 0.00082$	$R^2 = 0.00171$	$R^2 = 0.00201$	$R^2 = 0.00174$	$R^2 = 0.00166$
	0.716	0.102	0.050	0.083	-0.322
Sep	(0.438)	(0.724)	(0.712)	(0.778)	(0.583)
	$R^2 = 0.00028$	$R^2 = 0.00006$	$R^2 = 0.00006$	$R^2 = 0.00004$	$R^2 = 0.00014$
	2.040	0.250	0.201	0.270	0.664
Oct	(0.254)	(0.478)	(0.517)	(0.345)	(0.275)
	$R^2 = 0.00057$	$R^2 = 0.00022$	$R^2 = 0.00018$	$R^2 = 0.00039$	$R^2 = 0.00047$
	0.528	0.147	0.292	0.052	0.524
Nov	(0.748)	(0.525)	(0.205)	(0.604)	(0.402)
	$R^2 = 0.00005$	$R^2 = 0.00019$	$R^2 = 0.00072$	$R^2 = 0.00012$	$R^2 = 0.00031$
	0.033	-0.106	0.008	0.192	0.316
Dec	(0.528)	(0.615)	(0.781)	(0.491)	(0.540)
	$R^2 = 0.00017$	$R^2 = 0.00011$	$R^2 = 0.00003$	$R^2 = 0.00021$	$R^2 = 0.00015$

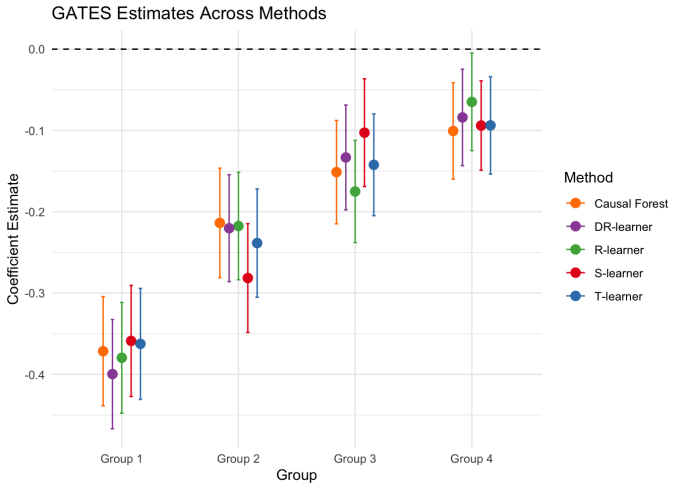
Table 6: Coefficients β_2 with their estimated median p-value across 100 A-M splits for the BLP considering different estimators of the CATE. The same analysis has been done with a single split without significant changes. P-values are clustered at the individual level.

	Group	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
S-Learner	**1**	0.039	0.001	-0.008	-0.346	-0.220	-0.136	-0.069	-0.072	-0.059	-0.051	-0.028	-0.017
		(0.298)	(0.318)	(0.671)	(0.000)	(0.000)	(0.000)	(0.050)	(0.046)	(0.093)	(0.119)	(0.432)	(0.631)
	2	0.009	-0.010	-0.013	-0.193	-0.184	-0.081	-0.069	-0.056	-0.045	-0.018	0.010	0.003
		(0.343)	(0.570)	(0.537)	(0.000)	(0.000)	(0.018)	(0.028)	(0.076)	(0.162)	(0.548)	(0.744)	(0.524)
	3	-0.018	0.017	-0.001	-0.131	-0.134	-0.062	-0.046	-0.040	0.003	-0.016	-0.008	0.007
		(0.601)	(0.560)	(0.437)	(0.000)	(0.000)	(0.038)	(0.105)	(0.177)	(0.576)	(0.442)	(0.601)	(0.448)
	4	-0.003	-0.037	0.006	-0.094	-0.073	-0.053	-0.038	-0.032	-0.031	-0.008	-0.005	-0.011
		(0.604)	(0.262)	(0.425)	(0.002)	(0.009)	(0.107)	(0.254)	(0.357)	(0.335)	(0.804)	(0.584)	(0.515)
T-Learner	**1**	0.037	-0.016	-0.062	-0.381	-0.273	-0.125	-0.134	-0.113	-0.046	-0.042	-0.012	0.002
		(0.320)	(0.542)	(0.070)	(0.000)	(0.000)	(0.001)	(0.000)	(0.003)	(0.170)	(0.192)	(0.627)	(0.836)
	2	-0.008	0.002	0.022	-0.157	-0.167	-0.118	-0.038	-0.053	-0.022	-0.013	0.019	-0.014
		(0.441)	(0.723)	(0.335)	(0.000)	(0.000)	(0.001)	(0.221)	(0.107)	(0.493)	(0.647)	(0.448)	(0.537)
	3	-0.016	-0.001	0.001	-0.107	-0.104	-0.058	-0.005	-0.022	-0.039	-0.013	-0.013	-0.015
		(0.447)	(0.794)	(0.635)	(0.002)	(0.001)	(0.065)	(0.447)	(0.373)	(0.192)	(0.672)	(0.615)	(0.412)
	4	-0.023	0.020	0.033	-0.056	-0.046	-0.031	-0.035	-0.033	-0.034	0.004	-0.001	-0.019
		(0.423)	(0.308)	(0.325)	(0.054)	(0.066)	(0.297)	(0.242)	(0.303)	(0.336)	(0.402)	(0.753)	(0.576)
R-Learner	**1**	0.063	-0.008	-0.015	-0.346	-0.244	-0.137	-0.105	-0.088	-0.027	-0.032	-0.027	-0.003
		(0.083)	(0.786)	(0.391)	(0.000)	(0.000)	(0.000)	(0.003)	(0.016)	(0.441)	(0.307)	(0.208)	(0.805)
	2	-0.028	-0.011	-0.016	-0.136	-0.174	-0.109	-0.035	-0.045	-0.044	-0.032	-0.017	-0.008
		(0.391)	(0.692)	(0.401)	(0.000)	(0.000)	(0.002)	(0.291)	(0.155)	(0.159)	(0.274)	(0.542)	(0.807)
	3	-0.018	-0.009	-0.008	-0.111	-0.113	-0.067	-0.060	-0.029	-0.041	-0.004	-0.015	-0.006
		(0.432)	(0.765)	(0.611)	(0.001)	(0.000)	(0.044)	(0.049)	(0.325)	(0.187)	(0.873)	(0.503)	(0.353)
	4	-0.008	-0.009	0.028	-0.107	-0.057	-0.026	-0.025	-0.010	-0.050	-0.014	0.009	-0.018
		(0.620)	(0.584)	(0.409)	(0.001)	(0.035)	(0.404)	(0.442)	(0.624)	(0.169)	(0.450)	(0.658)	(0.569)
DR-Learner	**1**	0.037	-0.022	-0.031	-0.347	-0.289	-0.147	-0.085	-0.097	-0.015	-0.049	0.000	-0.029
		(0.226)	(0.542)	(0.360)	(0.000)	(0.000)	(0.000)	(0.018)	(0.008)	(0.613)	(0.148)	(0.533)	(0.366)
	2	-0.025	-0.027	0.008	-0.145	-0.150	-0.078	-0.064	-0.046	-0.030	-0.031	-0.006	-0.002
		(0.381)	(0.371)	(0.603)	(0.000)	(0.000)	(0.022)	(0.051)	(0.159)	(0.323)	(0.245)	(0.816)	(0.506)
	3	-0.005	0.001	0.009	-0.112	-0.121	-0.089	-0.046	-0.043	-0.082	-0.022	0.013	-0.003
		(0.613)	(0.591)	(0.159)	(0.000)	(0.000)	(0.005)	(0.127)	(0.164)	(0.010)	(0.494)	(0.558)	(0.924)
	4	0.011	0.013	0.005	-0.090	-0.074	-0.017	-0.060	-0.026	-0.035	0.019	-0.034	-0.017
		(0.525)	(0.609)	(0.657)	(0.002)	(0.006)	(0.532)	(0.051)	(0.427)	(0.289)	(0.517)	(0.333)	(0.620)
Causal Random Forests	**1**	0.021	-0.013	-0.049	-0.367	-0.286	-0.149	-0.099	-0.086	0.002	-0.028	-0.008	-0.016
		(0.540)	(0.500)	(0.160)	(0.000)	(0.000)	(0.000)	(0.006)	(0.021)	(0.534)	(0.356)	(0.684)	(0.641)
	2	0.008	0.006	-0.003	-0.144	-0.175	-0.096	-0.056	-0.024	-0.061	-0.026	-0.038	-0.047
		(0.602)	(0.701)	(0.722)	(0.000)	(0.000)	(0.007)	(0.088)	(0.481)	(0.050)	(0.409)	(0.255)	(0.167)
	3	-0.000	-0.026	-0.002	-0.127	-0.125	-0.044	-0.023	-0.059	-0.032	-0.023	-0.000	0.017
		(0.452)	(0.446)	(0.476)	(0.000)	(0.000)	(0.180)	(0.445)	(0.058)	(0.329)	(0.463)	(0.834)	(0.571)
	4	0.019	0.016	0.008	-0.070	-0.040	-0.017	-0.035	0.000	-0.023	0.009	0.020	-0.007
		(0.511)	(0.478)	(0.804)	(0.019)	(0.106)	(0.562)	(0.249)	(0.600)	(0.346)	(0.570)	(0.521)	(0.573)

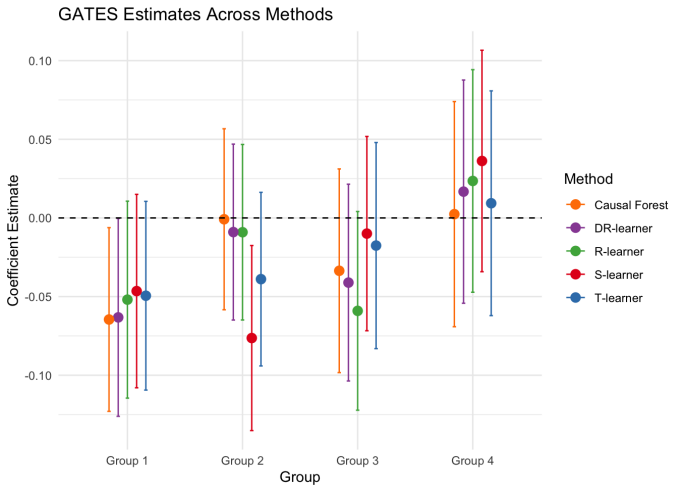
Table 7: Median GATES coefficients with their estimated median p-value across 100 A-M splits for the GATEs considering different estimators of CATE. The same analysis was performed with a single split with no significant changes. The results are available on request.



January



April



October

Figure 9: GATES estimates from January, April, and October 2020. The results are shown for the four quartiles according to CATE. In each graph, the colored bars are from left to right for Causal Forest (orange), DR-learner (purple), R-learner (green), S-learner (red) and T-learner (blue). Figure Appx.1 in Appendix B shows the estimates for all months of 2020.

Having the BLP and GATES analyses confirmed the substantial heterogeneity detected by the SPE, it becomes particularly interesting to further explore the sources of these differences following a procedure that is exactly the same of the one used in the previous CA analyses. What is different here is the inference part. We repeat 100 times the split in A and M samples, together with the calculation of the first and third quartiles of the distribution of the estimated CATE and the comparison of the mean characteristics of the firms contained in the tails of the distribution. Then we aggregate the results by taking the median of the estimated difference in means and the corresponding p-values. More formally, let X_{t_s-1} denote the vector of observed covariates. The CLAN compares the average covariate values between the "least affected group" G_1 and the "most affected group" G_K, as defined by the GATES framework.

Table 8 presents the results of the CLAN across various estimators. The results confirm the findings from the CA analysis reported in Table 5, offering an additional diagnostic.

The CLANs in Table 8 highlights a consistent pattern of treatment effect heterogeneity across industrial characteristics, export behavior, and time dummies. For instance, the Textile sector shows systematically higher representation among more affected firms, especially under the R- and DR-Learners, suggesting that sectoral affiliation plays a key role in shaping heterogeneity. This is broadly aligned with the CA findings, where Textile also shows positive and statistically significant estimate. Temporal patterns are also consistent across methods. Months corresponding to the COVID-19 peak (April–June) exhibit strong and positive CLAN differences, particularly under the Causal Random Forest. Differences in trade intensity are particularly stark. CLAN reveals substantial negative differences in Value Exported (log) and Value Imported (log) across all estimators. These results suggest that more severely affected firms are systematically smaller traders and confirm the previous findings from CA. Moreover, similar to the case of diversification, analyzing both the export and import sides supports and reinforces the previous finding that greater experience in importing inputs from abroad reduces the severity of the shock.

	T-Learner			S-Learner			R-Learner			DR-Learner			Causal Random Forest		
Outcome variable	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
Agriculture	-0.0893			-0.0528			-0.0911			-0.0941			-0.0546		
	(0.0000)			(0.0000)			(0.0000)			(0.0000)			(0.0000)		
Chemicals	0.0011			-0.0268			-0.0181			-0.0155			-0.0287		
	(0.5286)			(0.0014)			(0.0796)			(0.2001)			(0.0011)		
Manufacturing	-0.0120			-0.0261			-0.0109			-0.0131			-0.0149		
	(0.3080)			(0.0014)			(0.4132)			(0.2889)			(0.1543)		
Metals	-0.0102			-0.0209			-0.0146			-0.0123			-0.0246		
	(0.5253)			(0.0059)			(0.1767)			(0.2889)			(0.0013)		
Special	0.0004			-0.0041			0.0008			0.0005			-0.0026		
	(1.0000)			(0.7961)			(1.0000)			(1.0000)			(1.0000)		
Textile	0.1045			0.0797			0.1035			0.1033			0.0873		
	(0.0000)			(0.0000)			(0.0000)			(0.0000)			(0.0000)		
Wood	0.0237			0.0086			0.0283			0.0285			0.0175		
	(0.0000)			(0.3938)			(0.0000)			(0.0000)			(0.0036)		
Air	0.1363	0.1153	0.2089	0.1084	0.0874	0.1938	0.1249	0.1011	0.1869	0.1270	0.1041	0.1973	0.1293	0.1053	0.3802
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Land	0.0119	0.0006	0.0076	-0.0159	-0.0140	-0.0337	-0.0001	-0.0040	-0.0007	0.0007	-0.0021	0.0002	-0.0221	-0.0151	-0.0284
	(0.1596)	(0.6053)	(0.5307)	(0.0806)	(0.1157)	(0.0137)	(1.0000)	(0.9917)	(0.4868)	(1.0000)	(1.0000)	(1.0000)	(0.0087)	(0.0684)	(0.0938)
Sea	-0.1553	-0.1278	-0.2486	-0.1422	-0.1168	-0.2833	-0.1716	-0.1414	-0.2809	-0.1682	-0.1345	-0.2914	-0.1661	-0.1353	-0.4264
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
January	-0.1563	-0.1564		-0.1505	-0.1509		-0.1532	-0.1545		-0.1524	-0.1546		-0.2026	-0.2036	
	(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)	
February	-0.1545	-0.1567		-0.1534	-0.1545		-0.1760	-0.1782		-0.1736	-0.1765		-0.2229	-0.2263	
	(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)	
March	-0.1304	-0.1138		–0.1457	–0.1237		-0.1302	-0.1109		-0.1342	-0.1400		-0.1771	-0.1680	
	(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)	
April	0.3136	0.3192		0.3487	0.3552		0.3211	0.3302		0.3247	0.3340		0.3620	0.3679	
	(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)	
May	0.2975	0.2983		0.3268	0.3292		0.3021	0.3024		0.3029	0.3040		0.3504	0.3527	
	(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)	
June	0.1276	0.1258		0.1239	0.1217		0.1285	0.1267		0.1262	0.1245		0.1779	0.1746	
	(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)	
July	0.0106	0.0097		-0.0129	-0.0140		0.0143	0.0129		0.0137	0.0121		0.0296	0.0279	
	(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)	
August	-0.0203	-0.0199		-0.0370	-0.0384		-0.0180	-0.0187		-0.0245	-0.0260		-0.0125	-0.0135	
	(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)	
September	-0.0741	-0.0725		-0.1040	-0.1027		-0.0790	-0.0778		-0.0781	-0.0757		-0.0799	-0.0784	
	(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)	
October	-0.0893	–0.0908		-0.0994	-0.1010		-0.0877	-0.0906		-0.0950	-0.0964		-0.0995	-0.1009	
	(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)	
November	-0.1270	-0.1295		-0.1305	-0.1308		-0.1337	-0.1356		-0.1354	-0.1379		-0.1658	-0.1644	
	(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)	
December	-0.1238	-0.1243		-0.1213	-0.1213		-0.1278	-0.1270		-0.1271	-0.1263		-0.1577	-0.1557	
	(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)	
Number of export destination (ND)	-0.3441	-0.2736	-0.4438	-0.5962	-0.5076	-1.2120	-0.5040	-0.4199	-0.6698	-0.5007	-0.3931	-0.6800	-0.4773	-0.3900	-1.2712
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Number of import origins (NO)	-0.7602	-0.7267	-1.2371	-1.0417	-0.9062	-1.8311	-0.8403	-0.7905	-1.3375	-0.8417	-0.8095	-1.3305	-1.0015	-0.8843	-2.2725
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Number of exported products (NP)	0.0960	-0.2116	-0.2787	-0.3858	-0.3373	-0.6704	-0.0410	-0.2689	-0.2510	-0.0393	-0.2542	-0.2988	-0.2322	-0.2975	-0.3978
	(0.6289)	(0.0343)	(0.1162)	(0.0011)	(0.0000)	(0.0000)	(0.5352)	(0.0108)	(0.0921)	(1.0000)	(0.0154)	(0.1211)	(0.2692)	(0.0050)	(0.0938)
Value Exported (log)	-0.534	-0.4667	-0.9113	-0.6138	-0.5327	-1.3369	-0.6400	-0.5569	-1.0672	-0.6160	-0.5401	-1.0956	-0.6873	-0.6020	-1.9425
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Value Imported (log)	-1.1318	-1.1975	-2.0821	-1.6113	-1.5225	-3.3357	-1.3843	-1.3448	-2.5116	-1.3679	-1.3148	-2.5215	-1.5995	-1.5205	-3.3083
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Deviation from sectoral mean		✓	✓		✓	✓		✓	✓		✓	✓		✓	✓
Deviation from monthly mean			✓			✓			✓			✓			✓

Table 8: CLANs with 100 splits for S-Learner, T-Learner, R-Learner, DR-Learner, Causal Random Forest. We report the median of joint p-values.

5 Concluding discussion

In this paper, we show the potential of ML techniques for building counterfactuals, identifying the most affected subpopulations and the sources of treatment effect heterogeneity in scenarios where a credible control group is unavailable and it is difficult to define ex-ante the intensity of the shock for each unit.

In the application we consider, we concentrate on the effects of an economy-wide shock

such as COVID-19 on a firm's export behaviour. Using data from the Colombian customs office, we estimate that, during 2020, on average, the COVID-19 shock decreased a firm's probability of surviving in the export market by about 15 to 20 percentage points in April and May and by approximately 5 to 8 percentage points in June and July. By analysing the estimated treatment effect distribution, we reveal that these average results hide considerable heterogeneity. For example, in April 2020, we find that for some exporters COVID-19 decreased their survival probability by 55 percentage points. We identify the firms most and least affected by COVID-19 and we compare their characteristics by integrating the Sorted Partial Effects methodology with our causal ML approach. We emphasize how the integration into global value chains on the import side, both in terms of the number of countries from which a firm sources and the value of imports, is an important factor of resilience for exporters facing the COVID-19 shock.

From a methodological point of view, we show practitioners how to apply the generic ML tools proposed by Chernozhukov et al. (2023) to a context in which there is no control group available; we suggest how to use in-time placebo tests to check the credibility of counterfactual estimates; finally, we provide evidence indicating that in the Sorted Partial Effects analysis, in which the focus lies on the tails of the distribution of the treatment effects, it is critical to correct the estimation error arising from the necessarily imperfect reconstruction of the unobservable counterfactual.

While this method is specifically designed for analyzing the heterogeneous impacts of economy-wide shocks, there exists potential utility in employing this approach in less extreme situations where policies or shocks may exhibit unobservable spillovers that are challenging to model in advance. In such contexts, our empirical framework proves advantageous in detecting these potential heterogeneous indirect effects, as it circumvents the need for a priori identification of a control group of untreated units.

Finally, in this paper we also demonstrate that ML methods can be applied successfully to predict firms' trade potential. We consider ML methods a promising tool to assist firms and public agencies in their export decision-making processes. The bulk of countries possesses export promotion agencies whose objective is to sustain firms' internationalization activities by lowering the costs of information acquisition (Broocks and Van Biesebroeck, 2017; Munch and Schaur, 2018). Given that exporters' dynamics can be understood as a complex learning process dense of interdependencies (complementarity or substitutability) between products and destination markets (from the perspective of technology/knowledge, local tastes, legal requirements, and marketing and distribution costs) and that ML techniques have been successfully applied to predict firm performances in such settings, we believe that an important avenue for future research is to test the effectiveness of using these techniques and firm-level data to build recommendation systems. These systems could help firms identify their latent comparative advantages and provide export diversification and differentiation recommendations.

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A Appendix - Data

Table Appx.1: Predictors for exporters' success

Variable	Description	Source
	Models: SUM and SAM	
NP, ND, NO	Number of products exported by, number of destinations where a company exports, and number of import origin countries for an exporter in a given month, respectively.	Authors' own elaboration from Colombian Customs Office (DIAN).
HH_p, HH_d	Product-Herfindahl Index, and Destination-Herfindahl Index. Measure the concentration of products at 6-digit HS, and the concentration at destination by company-month, respectively.	Authors' own elaboration from the Colombian Customs Office (DIAN).
Total value (exports)	Free on board value of the export transaction in US dollars for each company-month.	Colombian Customs Office (DIAN)
Total value (imports)	Free on board value of the import transaction in US dollars for each company-month.	Colombian Customs Office (DIAN)
Size	4 class dummies classifying firms according to the quartiles of the firm-level (Q1, Q2, Q3 and Q4) distribution of the total monthly value of exports (in ln).	Authors' own elaboration
Destination	Factor variable with one level (dummy variable) for each destination country where Colombian exporters operate by month.	Colombian Customs Office (DIAN)
Origin	Factor variable with one level (dummy variable) for each import origin country, where Colombian exporters operate by month.	Colombian Customs Office (DIAN)
Continent	Factor variable with one level (dummy variable) for each continent where Colombian exporters operate.	Authors' own elaboration
Department	Factor variable with one level (dummy variable) for each department (region) in Colombia from which companies operate.	Colombian Customs Office (DIAN)
Means of Transportation	4 class dummies indicating the means of transportation a company uses to perform a transaction (land, sea, air, others).	Colombian Customs Office (DIAN)
Sector	99 class dummies for the classification of company products by 2-digit HS code (corresponding to HS-chapters).	Authors' own elaboration
Industry	22 class dummies indicating the industries (HS-sections) where companies operate.	Authors' own elaboration from the Colombian Customs Office (DIAN).
Sector Experience	Factor variable with one level (dummy variable) for each sector. Takes value 1 in all periods after a company exports for the first time in a given sector (reflecting past experience in a sector).	Authors' own elaboration from the Colombian Customs Office (DIAN).
Destination Experience	Factor variable with one level (dummy variable) for each destination. Takes value 1 in all periods after a company exports for the first time in a given destination (reflecting past experience in a destination).	Authors' own elaboration from the Colombian Customs Office (DIAN).
Exporter (importer) Experience	Variable counting the accumulated value exported (imported) in the last twelve months.	Authors' own elaboration from the Colombian Customs Office (DIAN).
	Models: SAM (COVID-19 variables)	
Containment Economic Index	We consider the Economic Index from Hale et al. (2020) that provides a measure of the strength of the economic policies set in place to deal with the pandemic (such as income support and debt relief) for each country in the world. It ranges from 0 to 100. At the firm level we define two variables, one at the export and one at the import side, by taking a weighted average of these country level scores according to the monthly value of exports(imports) that a company ships(source) in every country.[1]	Hale et al. (2020) and Colombian Customs Office (DIAN).
Containment Government Index	We consider the Government Index from Hale et al. (2020) that measures the strictness of 'lockdown' style policies that primarily restrict people's behavior. It ranges from 0 to 100. At the firm level we define two variables, one at the export and one at the import side, by taking a weighted average of these country level scores according to the monthly value of exports(imports) that a company ships(source) in every country.	Hale et al. (2020) and Colombian Customs Office (DIAN).
Containment Health Index	We consider the Health Index from Hale et al. (2020) that combines 'lockdown' restrictions and closures with measures such as testing policy and contact tracing, short-term investment in healthcare, as well as investments in vaccine. Ranges from 0 to 100. At the firm level, we define two variables, one at the export and one at the import side, by taking a weighted average of these country-level scores according to the monthly value of exports(imports) that a company ships (source) in every country.	Hale et al. (2020) and Colombian Customs Office (DIAN).
Containment Stringency Index	We consider the Stringency Index from Hale et al. (2020) that records how the response of governments has varied over all indicators, becoming stronger or weaker over the course of the outbreak. Ranges from 0 to 100. At the firm level we define two variables, one at the export and one at the import side, by taking a weighted average of these country level scores according to the monthly value of exports(imports) that a company ships(source) in every country.	Hale et al. (2020) and Colombian Customs Office (DIAN).
	Models: SUM and SAM (variables only for Logit, Logit-LASSO, Logit-Ridge and SVM)	
Size*Industry	Factor variables with 5 levels for each industry. It takes the value 1 if the firm size is Q1, the value 2 if the firm size is Q2, the value 3 if the size is Q3 and the value 4 if the size is Q4 while operating in a particular industry. However, it takes the value 0 if a company is not active in this industry (regardless of size).	Authors' own elaboration based on the Colombian Customs Office (DIAN).
Size*Sector	Factor variables with 5 levels for each sector. It takes the value 1 if the firm size is Q1, the value 2 if the firm size is Q2, the value 3 if the size is Q3 and the value 4 if the size is Q4 while operating in a specific sector. However, it takes the value 0 if a company is not active in this sector (regardless of size).	Authors' own elaboration based on the Colombian Customs Office (DIAN).
Size*Means of Transportation	Factor variables with 5 levels for each sector. It takes the value 1 when the company size is Q1, value 2 when the company size is Q2, value 3 when the size is Q3, and value 4 when the size is Q4 while operating using a given means of transportation. However, it takes value 0 if a company is not operating using this means of transportation (for any size level).	Authors' own elaboration from the Colombian Customs Office (DIAN).
Size*Destination	Factor variables with 5 levels for each sector. It takes the value 1 when the company size is Q1, value 2 when the company size is Q2, value 3 when the size is Q3, and value 4 when the size is Q4 while operating in a given destination. However, it takes value 0 if a company is not operating in this destination (for any size level).	Authors' own elaboration from the Colombian Customs Office (DIAN).

* https://data.europa.eu/euodp/en/data/dataset/covid-19-coronavirus-data

[1] When an exporter does not import, we impute the corresponding internal Index (Economic, Government, Health, and Stringency) of Colombia to create the corresponding import side Index.

[2] Only the variables and interactions listed in this table were used in the analysis (no second or higher degree polynomial function). Interactions were removed in tree-based models.

Table Appx.2: Sector-Industry mapping

Section (Industry)	Industry Name	HS-Chapter (Sector)
1	Live Animals/ Animal Products	1-5
2	Vegetable Products	6-14
3	Animal or Vegetable Fats/Oils	15
4	Prepared Foodstuffs	16-24
5	Mineral Products	25-27
6	Products of Chemical Industries	28-38
7	Plastics, Rubber	39-40
8	Raw Hides, Skins and Leather	41-43
9	Wood	44-46
10	Paper	47-49
11	Textile	50-63
12	Footwear	64-67
13	Art. of Stone, Cement	68-70
14	Jewelries	71
15	Base Metals	72-83
16	Machinery Equipment	84-85
17	Vehicles	86-89
18	Precision Instruments	90-92
19	Arms	93
20	Miscellaneous Manufactured Articles	94-96
21	Works of Art	97
22	Special Classification Provisions	98-99

Source: Author's elaboration using Pierce and Schott (2012) tables.

B An overview of the alternative CATE estimators

The BLP, GATEs and CLAN analysis we presented has been performed by using the CATEs estimated by using various meta-learners and the *Causal Random Forests* (CFs) method. CFs modifies a particular standard (predictive) machine learning method (Random Forests) so that it directly targets the estimation of the CATE. Instead, meta-learners operate through a multi-step procedure that breaks down the task of estimating CATE into several smaller sub-problems that can be addressed using any standard (predictive) machine learning technique. They typically involve the following steps:

1. Estimation of nuisance parameters: auxiliary components such as the propensity scores are estimated using machine learning algorithms.

2. Construction of an objective function: The estimated nuisance components are then used to construct a tailored minimization problem whose solution targets the CATE function. This step is designed to isolate the heterogeneity in treatment effects while accounting for confounding.

3. Solution via machine learning: The resulting minimization problem is solved via machine learning.

4. Prediction of CATEs: Finally, the learned model is used to generate predictions of the CATE for each observational unit, thus enabling individualized causal effect estimation.

The meta-learners we have used are the S-Learner, the T-Learner, the R-Learner and the DR-Learner. They are all based on the assumption of strong ignorability. We will provide a brief overview in what follows. $m(1,x)$ denotes the conditional mean function under treatment, $m(0,x)$ the conditional mean function in absence of treatment, and $e(X)$ the propensity score.
The first meta-learner we explore is the *S-Learner*. It fits a single model in which the observed outcome is modeled as a function of the covariates and the treatment indicator variable. The resulting model is then used to obtain two predictions for each subject: under treatment and control. The CATE is then estimated by taking the differences between the two predictions: $\hat{\Delta}^s(x) = \hat{m}(1,x) - \hat{m}(0,x)$. Instead, the T-Learner employs two different models estimated separately on the treated and control samples, and then the CATE is obtained as a difference, as in the S-Learner.

The *R-Learner*, introduced by Nie and Wager (2021), builds on the partially linear model while allowing for covariate-specific treatment effects. In this setting, the potential outcome model is given by $Y^D = \Delta(X)D + g(X) + U^D, \quad \text{with } \mathbf{E}[U^D \mid D, X] = 0$, which implies that the observed outcome satisfies $Y = D\Delta(X) + g(X) + U^D$. By defining the outcome regression function $m(X) := \mathbf{E}[Y \mid X]$, the model can be transformed into a residualized form:

$$Y - m(X) = \Delta(X)(D - e(X)) + U^D.$$

This motivates the R-Learner objective:

$$\hat{\Delta}^{rl}(X) = \arg\min_{\Delta} \sum_{i=1}^{N} \left(Y_i - \hat{m}(X_i) - \Delta(X_i)(W_i - \hat{e}(X_i))\right)^2,$$

where the nuisance parameters $\hat{m}(X)$ and $\hat{e}(X)$ are estimated using high-quality machine learning methods, typically via cross-fitting [33] (Nie and Wager, 2021).
When one does not wish to impose linearity on $\Delta(X)$, the R-Learner objective can be rewritten as

$$\hat{\Delta}^{rl}(X) = \arg\min_{\Delta} \sum_{i=1}^{N} (D_i - \hat{e}(X_i))^2 \left(\frac{Y_i - \hat{m}(X_i)}{D_i - \hat{e}(X_i)} - \Delta(X_i)\right)^2.$$

[33] A simple case arises, for instance, if we model the CATE as a linear function $\Delta(X) = X'\beta$, the minimization becomes

$$\hat{\beta}^{rl} = \arg\min_{\beta} \sum_{i=1}^{N} \left(Y_i - \hat{m}(X_i) - \tilde{X}_i'\beta\right)^2,$$

where $\tilde{X}_i = (D_i - \hat{e}(X_i))X_i$ are the so-called modified or pseudo-covariates. The estimated CATE is then given by $\hat{\Delta}^{rl}(x) = x'\hat{\beta}^{rl}$, noting that this differs from $\tilde{X}\hat{\beta}^{rl}$. In this case, obtaining $\hat{\beta}^{rl}$ reduces to a standard regression of the residualized outcome on the modified covariates, and shrinkage estimators such as Lasso can be readily applied. Importantly, the nuisance parameters can still be estimated using non-linear machine learning methods

This representation highlights that any supervised learning algorithm capable of handling weighted minimization problems can be employed (e.g. neural networks, random forests, and gradient boosting among others). In this formulation, the weights are $(D_i - \hat{e}(X_i))^2$, the pseudo-outcome is $\frac{Y_i - \hat{m}(X_i)}{D_i - \hat{e}(X_i)}$, and the original covariates X are used as features.

The *DR-Learner*, introduced by Kennedy et al. (2020), constructs a pseudo-outcome in the first stage, defined as

$$\tilde{Y}_{\text{ATE}} = \underbrace{\hat{m}(1, X) - \hat{m}(0, X)}_{\text{outcome predictions}} + \underbrace{\frac{D(Y - \hat{m}(1, X))}{\hat{e}(X)} - \frac{(1 - D)(Y - \hat{m}(0, X))}{1 - \hat{e}(X)}}_{\text{weighted residuals}},$$

similarly to Eq. (15) and targets the CATE function through the conditional expectation

$$\Delta(x) = \mathbf{E}\left[\tilde{Y}_{\text{ATE}} \mid X = x\right].$$

Since $\mathbf{E}[\tilde{Y}_{\text{ATE}} \mid X]$ is a conditional expectation function of random variable, it can be approximated using standard supervised learning techniques. The DR-Learner uses this pseudo-outcome as the dependent variable in a generic machine learning regression:

$$\hat{\Delta}^{\text{dr}}(X) = \arg\min_{\Delta} \sum_{i=1}^{N} \left(\tilde{Y}_{i,\text{ATE}} - \Delta(X_i)\right)^2.$$

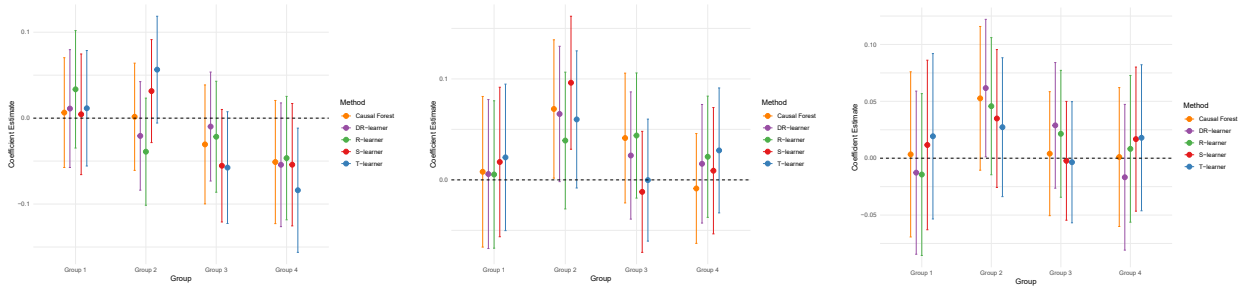
The *Causal Random Forests* (CFs) estimator cannot be properly considered a Meta-learner because it alters a specific ML method in such a way that it estimates the CATE. The development of *Causal Random Forests* (CFs) has evolved through multiple stages. It was introduced by Wager and Athey (2018) primarily for experimental settings. Subsequently, Athey et al. (2019) extended this approach by proposing an approximation to the splitting rule of Causal Trees for binary random treatments, while also generalizing the method to observational settings and continuous treatments. Conceptually, modern Causal Random Forests estimate CATEs via a localized, individualized residual-on-residual regression of the form

$$\hat{\Delta}^{\text{cf}}(x) = \arg\min_{\Delta} \left\{ \sum_{i=1}^{N} \pi_i(x) \left[(Y_i - \hat{m}(X_i)) - \Delta(x)(D_i - \hat{e}(X_i))\right]^2 \right\},$$

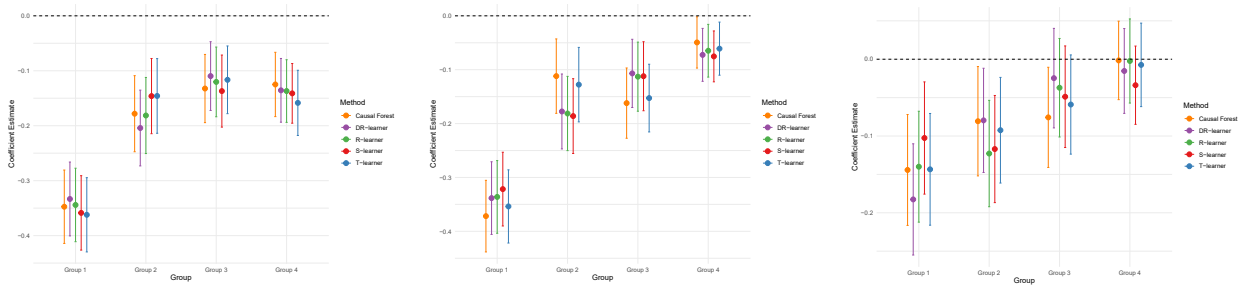
where $\pi_i(x)$ denotes the frequency with which the i-th training sample falls into the same leaf as the target sample x. This procedure represents a localized version of the partially linear estimator, with the nuisance components $\hat{m}(X)$ and $\hat{e}(X)$ being estimated in a preliminary step, typically through cross-fitting.

Finally, in the main text, for comparability with CFs, we use Random Forest in steps 1 and 3 of the meta-learners described above.

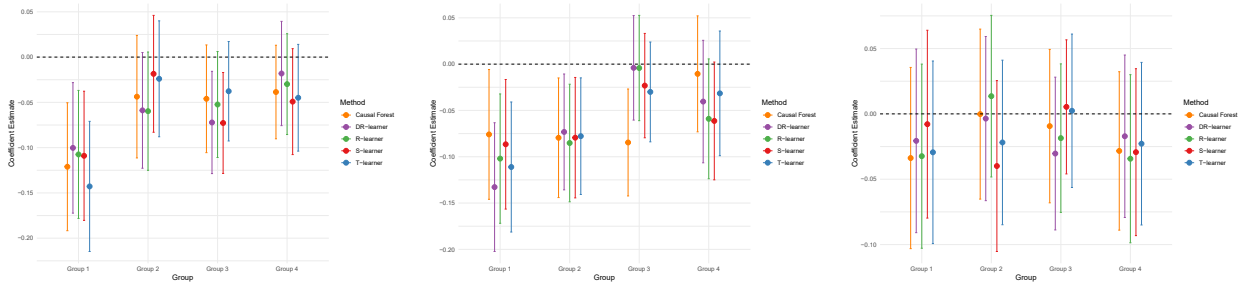
GATES estimates from January to December 2020 are summarized in Figure Appx.1.



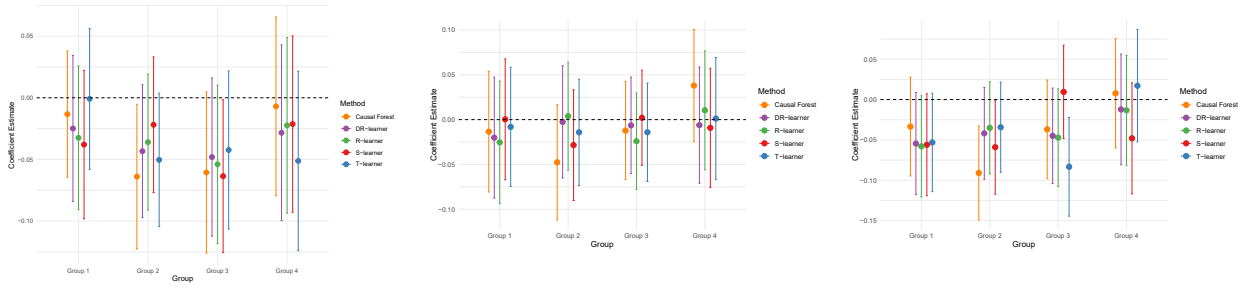
January February March



April May June



July August September



October November December

Figure Appx.1: GATES estimates from January to December 2020. The results are shown for the four quartiles according to CATE. In each graph, the colored bars are from left to right for Causal Forest (orange), DR-learner (purple), R-learner (green), S-learner (red) and T-learner (blue).

C DML AIPW estimator

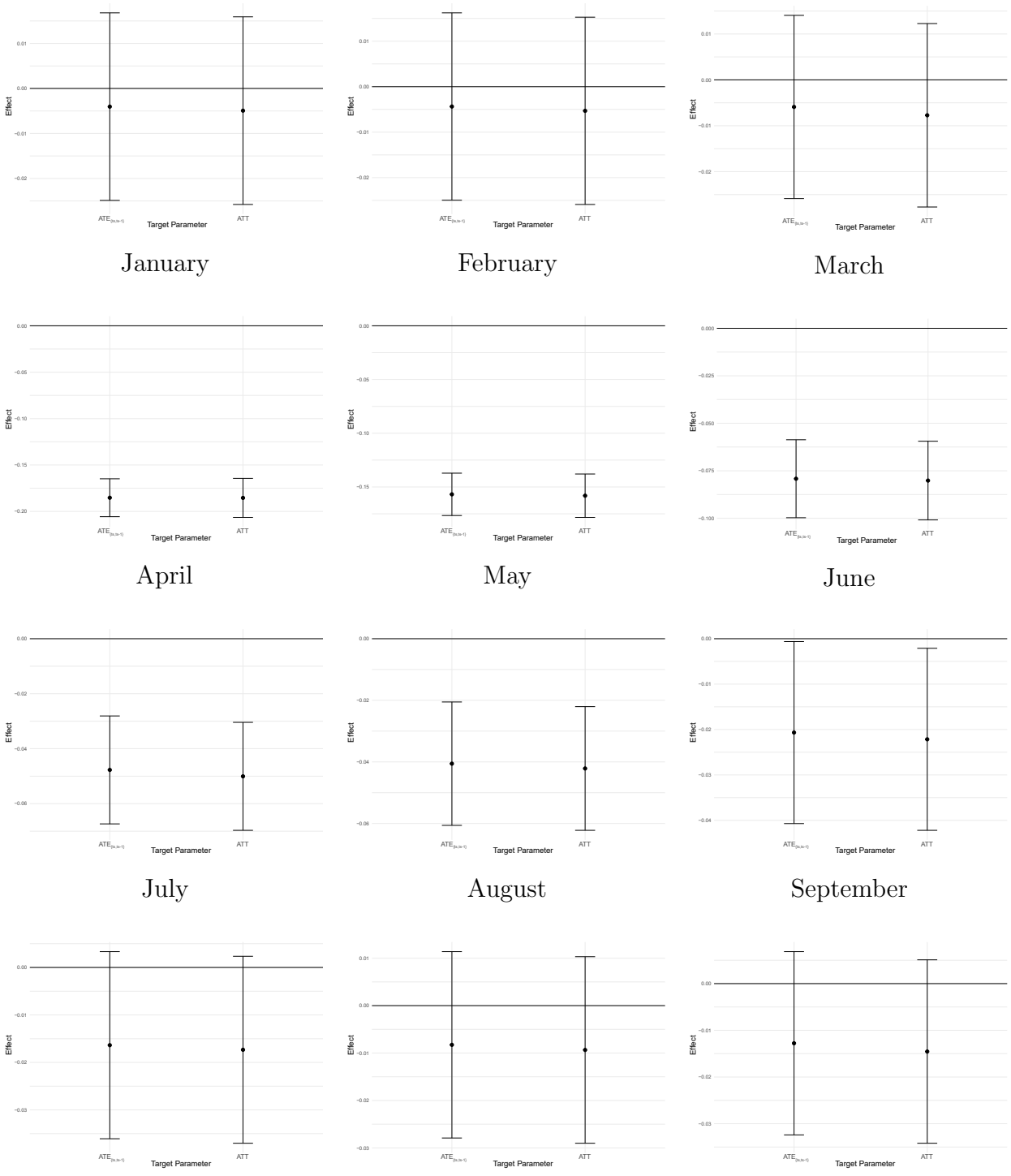


Figure Appx.2: Double Machine Learning (DML) AIPW estimates (with 5-fold cross-fitting and nuisance parameters estimated with Random Forest) from January to December 2020. $ATE_{\{t_s,t_{s-1}\}}$ refers to the ATE estimated by considering the combined cohort of treated (firms in t_s) and control firms (firms in t_{s-1}) as a unique sample. Notice also that $ATT = ATT_{\{t_s,t_{s-1}\}}$ because all the treated units are in t_s.

D Estimated Propensity Scores

The propensity score is defined in equation (3) as $P(D_{i,\{t_s,t_s-1\}} = 1|X_{i,\{t_s-1,t_s-2\}}) = e(X_{i,\{t_s-1,t_s-2\}})$, where $D_{i,\{t_s,t_s-1\}}$ is a dummy variable indicating whether an observation belongs to the treated group or to the control group, and $X_{i,\{t_s-1,t_s-2\}}$ are the corresponding explanatory variables. Therefore, the propensity score refers to the conditional probability of belonging to the cohort of firms observed in t_s considering the unique sample that combines the cohort of treated (firms observed in t_s) and the cohort of control firms (firms observed in $t_s - 1$). Following the methodology of Chernozhukov et al. (2018), a K-fold cross-fitting strategy is employed to estimate this quantity without overfitting and Random Forest is used.

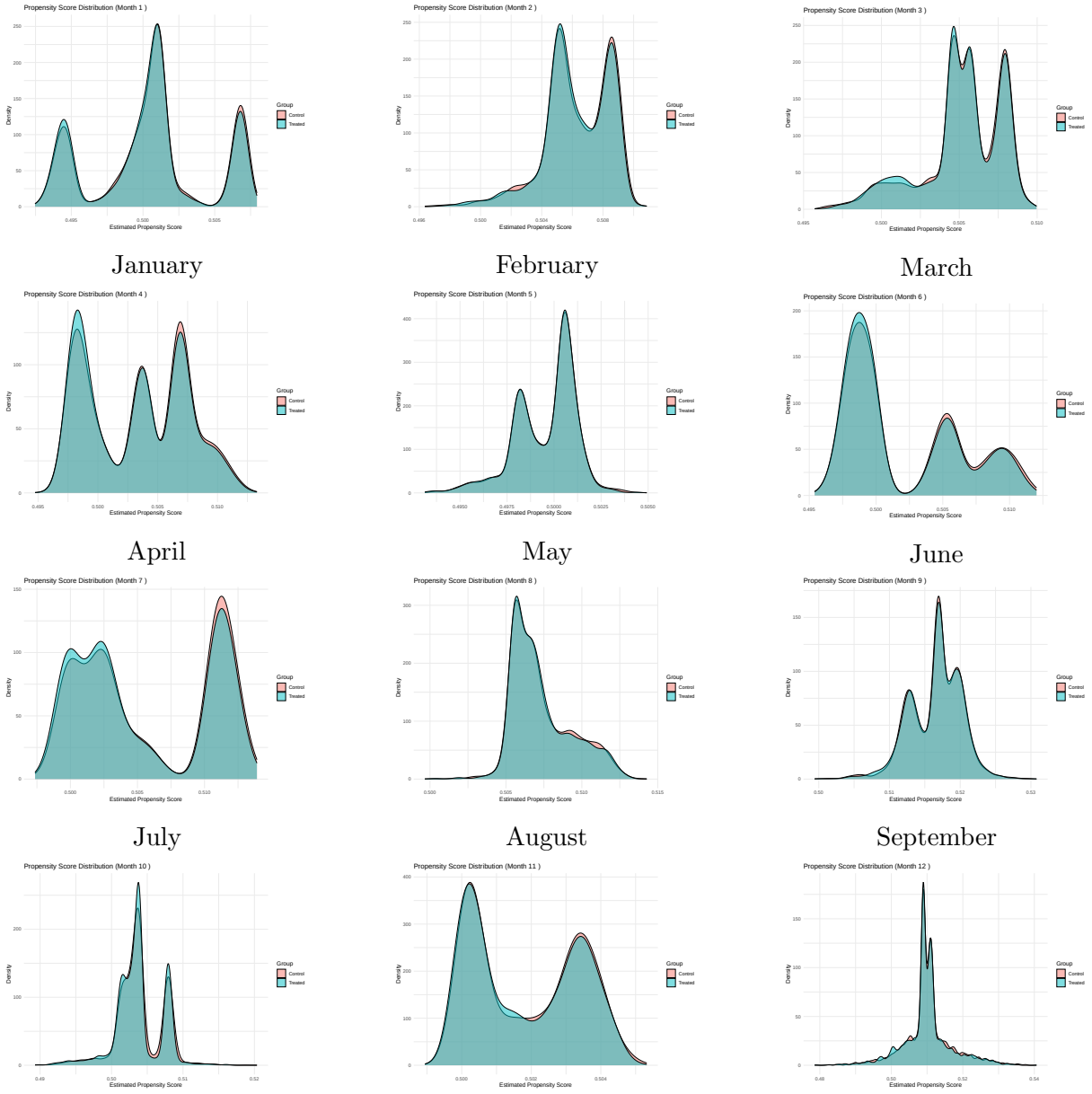


Figure Appx.3: Propensity scores estimates from January to December using 5-fold DML-AIPW based on Random Forest.